Maintenance and Service Guide

SUMMARY

This guide provides information about spare parts, removal and replacement of parts, security, backing up, and more.

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Product notice

This guide describes features that are common to most models. Some features may not be available on your computer.

Not all features are available in all editions or versions of Windows. Systems may require upgraded and/or separately purchased hardware, drivers, software or BIOS update to take full advantage of Windows functionality. Windows is automatically updated, which is always enabled. High-speed internet and Microsoft account required. ISP fees may apply and additional requirements may apply over time for updates. See http://www.microsoft.com. **If your product ships with Windows in S Mode:** Windows in S Mode works exclusively with apps from the Microsoft Store within Windows. Certain default settings, features, and apps cannot be changed. Some accessories and apps that are compatible with Windows may not work (including some antivirus, PDF writers, driver utilities, and accessibility apps), and performance may vary, even if you switch out of S Mode. If you switch to Windows, you cannot switch back to S Mode. Learn more at Windows.com/SmodeFAQ.

To access the latest user guides, go to http://www.hp.com/support, and follow the instructions to find your product. Then select **Manuals**.

Software terms

By installing, copying, downloading, or otherwise using any software product preinstalled on this computer, you agree to be bound by the terms of the HP End User License Agreement (EULA). If you do not accept these license terms, your sole remedy is to return the entire unused product (hardware and software) within 14 days for a full refund subject to the refund policy of your seller.

For any further information or to request a full refund of the price of the computer, please contact your seller.

Safety warning notice

Reduce the possibility of heat-related injuries or of overheating the computer by following the practices described.

⚠ **WARNING!** To reduce the possibility of heat-related injuries or of overheating the computer, do not place the computer directly on your lap or obstruct the computer air vents. Use the computer only on a hard, flat surface. Do not allow another hard surface, such as an adjoining optional printer, or a soft surface, such as pillows or rugs or clothing, to block airflow. Also, do not allow the AC adapter to come into contact with the skin or a soft surface, such as pillows or rugs or clothing, during operation. The computer and the AC adapter comply with the user-accessible surface temperature limits defined by applicable safety standards.

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1 Product description

This table provides detailed product information.

Table 1-1 Product components and their descriptions

Category	Description
Product Name	**HP Dragonfly Folio 13.5 inch G3 2-in-1 Notebook PC**
Processors	**12th generation Intel® Core™ processors**
	Intel Core i7-1265U processor (1.8 GHz [max turbo frequency 4.8 GHz], 10 cores, 12 MB Intel Smart Cache)
	Intel Core i7-1255U processor (1.7 GHz [max turbo frequency 4.7 GHz], 10 cores, 12 MB Intel Smart Cache)
	Intel Core i5-1245U processor (1.2 GHz [max turbo frequency 4.4 GHz], 10 cores, 12 MB Intel Smart Cache)
	Intel Core i5-1235U processor (1.3 GHz [max turbo frequency 4.4 GHz], 10 cores, 12 MB Intel Smart Cache)
Graphics	**Intel Iris® Xe Graphics**
	Supports HD Decode, DX12, HDMI™ 2.0
	DirectX support: 12.1
	OpenGL support: 4.6
	OpenCL support: 3.0
	Displays supported: 4
	Multiformat codec engines: 2
Display	**34.3 cm (13.5 in), IPS, ultrawide viewing angle (UWVA), touch screen, Corning® Gorilla® Glass 7, HP Eye Ease**
	WUXGA+ (1920 × 1280), sRGB 100%, BrightView Touch, 400 nits
	WUXGA+, Antiglare Touch, sRGB 100%, 400 nits
	WUXGA+, BrightView Touch, HP Sure View Reflect integrated privacy screen, sRGB 100%, 1000 nits
	3k2k (3000 × 2000), OLED, BrightView, DCI-P3 100%, 400 nits
Memory	Onboard, non upgradable memory supporting up to 32 GB of RAM
	LPDDR5-6400, supports dual channel memory
	Supports the following configurations:
	• 32 GB
	• 16 GB
	• 8 GB

Table 1-1 **Product components and their descriptions (continued)**

Category	Description
Primary storage	**M.2 2280 solid-state drives, PCIe, NVMe**
	2 TB, three layer cell (TLC)
	1 TB, TLC
	512 GB, TLC
	512 GB, TLC, self-encrypting drive (SED), OPAL2
	512 GB, value
	256 GB, TLC
	256 GB, TLC, SED, OPAL2
Audio and video	Bang & Olufsen
	Quad stereo speakers
	Integrated dual top-edge microphones
	Discrete amplifiers
	Camera, IR (infrared), 8 MP, 100° ultrawide viewing angle, HP Camera Privacy Key
Wireless	**Wireless Local Area Network (WLAN)** (onboard, not removable)
	Intel AX211 Wi-Fi® 6E Bluetooth® 5.2 WLAN
	Intel AX211 Wi-Fi 6E Bluetooth 5.2 vPro WLAN
	- Miracast® support
	- Support for HP LAN-Wireless Protection (WLAN/LAN/WWAN switching)
	- Support for HP Connection Optimizer
	- Support for HP Extended Range Wireless LAN
	- Supports UEFI Wi-Fi
	- Supports Indonesia New Band
	- Supports Time Average Power for WLAN
	- Supports Dynamic Antenna gain (EU only)
	- Supports InTile
	- Supports BT Audio Offload for A2DP
	Wireless Wide Area Network (WWAN) (select products only)
	Intel 5000 5G Solution WWAN
	- eSIM on module
	- M.2 30 × 52 socket PCIe
	Intel XMM 7560 R + LTE-Advanced Pro WWAN (Cat 16)
	- eSIM on module
	- Supports Global Series
	- M.2 30 × 42 socket PCIe

Table 1-1 **Product components and their descriptions (continued)**

Category	Description
Ports	Audio-out (headphone)/audio-in (microphone) combo jack
	(2) USB Type-C® power connectors and Thunderbolt™ ports with HP Sleep and Charge
Sensors	Hall sensor
	Accelerometer + Gyroscope + eCompass
	Ambient light sensor
	Adaptive color sensor
	IR thermal sensor
	Magnetometer
	Time of Flight
Security	Trusted Platform Module (TPM) 2.0
	Infineon SLB9672VU2.0
	Version: 15.21
	FIPS 140-2 compliant
	IPv6 support
Keyboard/pointing devices	**HP Premium Keyboard**
	Backlit, spill resistant
	Backlit, spill resistant, privacy
	Clickpad
	Multitouch gesture support
Power requirements	**Battery**
	4 cell, 53 Whr
	HP Long Life
	HP Fast Charge Technology
	HP Smart AC adapters
	65 W, slim, nPFC, USB Type-C, 1.8 m (6 ft)
	65 W, standard, nPFC, USB Type-C, 1.8 m (6 ft)
	Power cord
	C5, 1.0 m (3.3 ft) premium
	C5, 1.0 m (3.3 ft) standard
Digital pen	HP Dragonfly Folio Pen (select products only)

Table 1-1 **Product components and their descriptions (continued)**

Category	Description
Operating system	Windows® 11 Pro
	Windows 11 Enterprise available with a Volume Licensing Agreement)
	Windows 11 Pro Education
	Windows 11 Home
	Windows 11 Home Single Language
	Windows 11 Pro
	Windows 10 Pro (available through downgrade rights from Windows 11 Pro)
	FreeDOS
Serviceability	**End user replaceable parts**
	AC adapter
	Pen

2 Components

Your computer features top-rated components. This chapter provides details about your components, where they are located, and how they work.

Right

Use the illustration and table to identify the components on the right side of the computer.

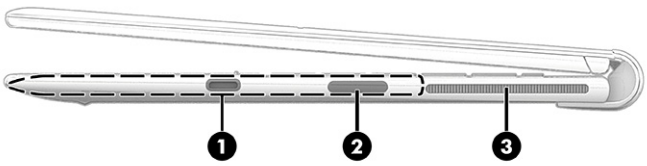


Table 2-1 Right-side components and their descriptions

Component		Description
(1)	Magnetic pen attachment area	Holds the magnetic pen against the side of the computer.
(2)	Pen charging area	Charges pen when attached to the computer in the magnetic pen attachment area. **NOTE:** You cannot remove the pen charging component from the outside of the computer.
(3)	Vent	Enables airflow to cool internal components. **NOTE:** The computer fan starts up automatically to cool internal components and prevent overheating. It is normal for the internal fan to cycle on and off during routine operation.

Left

Use the illustration and table to identify the components on the left side of the computer.

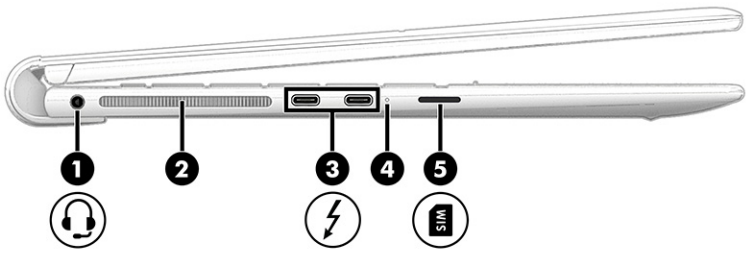


Table 2-2 Left-side components and their descriptions

Component		Description
(1)	Audio-out (headphone)/Audio-in (microphone) combo jack	Connects optional powered stereo speakers, headphones, earbuds, a headset, or a television audio cable. Also connects an optional headset microphone. This jack does not support optional standalone microphones. **WARNING!** To reduce the risk of personal injury, adjust the volume before putting on headphones, earbuds, or a headset. For additional safety information, see the *Regulatory, Safety, and Environmental Notices*. To access this guide: ■ Select the **Search** icon in the taskbar, type `HP Documentation` in the search box, and then select **HP Documentation**. **NOTE:** When a device is connected to the jack, the computer speakers are disabled.
(2)	Vent	Enables airflow to cool internal components. **NOTE:** The computer fan starts up automatically to cool internal components and prevent overheating. It is normal for the internal fan to cycle on and off during routine operation.
(3)	USB Type-C® power connectors and Thunderbolt™ ports with HP Sleep and Charge (2)	Connect an AC adapter that has a USB Type-C connector, supplying power to the computer and, if needed, charging the computer battery. – and – Connect a USB device, provides high-speed data transfer, and charges small devices (such as a smartphone), even when the computer is off. **NOTE:** Use a standard USB Type-C charging cable or cable adapter (purchased separately) when charging a small external device. **NOTE:** Your computer might also support a Thunderbolt docking station.
(4)	Battery light	When AC power is connected: • White: The battery charge is greater than 90 percent. • Amber: The battery charge is from 0 to 90 percent. • Off: The battery is not charging. When AC power is disconnected (battery not charging): • Blinking amber: The battery has reached a low battery level. When the battery has reached a critical battery level, the battery light begins blinking rapidly. • Off: The battery is not charging.
(5)	SIM card tray (select products only)	Provides access to the nano SIM card. For more information, see Using a SIM card (select products only) on page 14.

Display

Use the illustration and table to identify the display components.

Low blue light mode (select products only)

Your computer display is shipped from the factory in low blue light mode for improved eye comfort and safety. Also, blue light mode automatically adjusts blue light emissions when you are using the computer at night or for reading.

⚠ **WARNING!** To reduce the risk of serious injury, read the *Safety & Comfort Guide*. It describes proper workstation setup and proper posture, health, and work habits for computer users. The *Safety & Comfort Guide* also provides important electrical and mechanical safety information. The *Safety & Comfort Guide* is available on the web at http://www.hp.com/ergo.

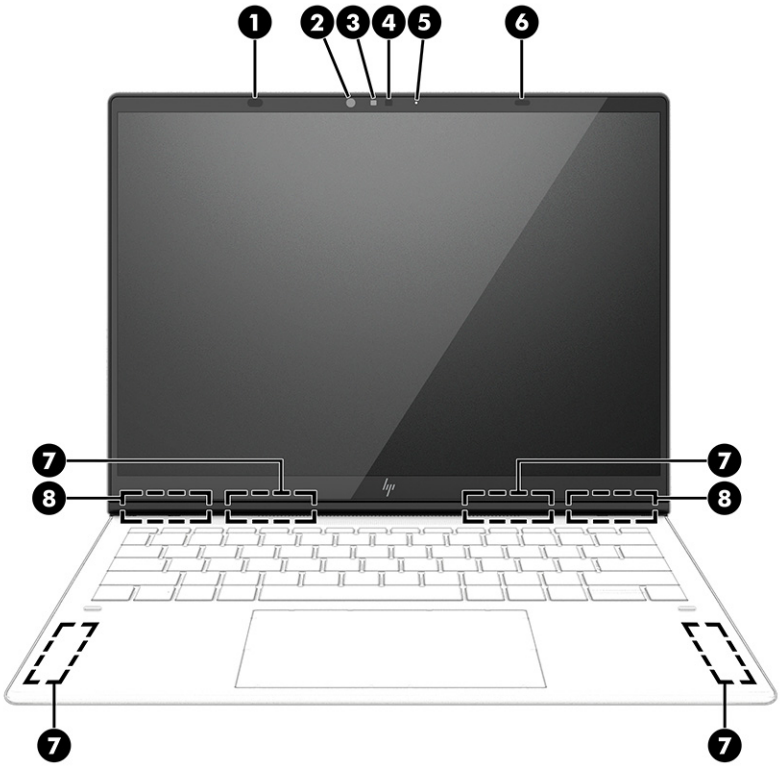


Table 2-3 Display components and their descriptions

Component		Description
(1)	User-proximity sensor	Uses facial recognition to monitor your presence in front of the computer to save power and add security. When you step away from the computer, the sensor puts the computer into the Sleep state. When you return to your computer, the sensor recognizes you and wakes the computer without a key press or mouse movement.
(2)	Infrared camera light	On: The camera is in use.
(3)	Infrared camera lens	Enables the use of the infrared features of the camera.

Table 2-3 **Display components and their descriptions (continued)**

Component		Description
(4)	Camera	Allows you to video chat, record video, and record still images. Some cameras also allow a facial recognition logon to Windows, instead of a password logon. **NOTE:** Camera functions vary depending on the camera hardware and software installed on your product.
(5)	Camera light	On: The camera is in use.
(6)	Ambient light and color sensor	Adjusts the brightness of the display, depending on the ambient light, and manages and controls the color temperature. Color temperature enables you to measure and adjust the warmth or coolness of the light source.
(7)	WWAN antennas*	Send and receive wireless signals to communicate with wireless wide area networks (WWANs).
(8)	WLAN antennas*	Send and receive wireless signals to communicate with wireless local area networks (WLANs).

*The ambient light, color, and other sensors, antennas, and some other components might be visible from the outside of the computer. For optimal transmission, keep the areas immediately around the antennas free from obstructions.

For wireless regulatory notices, see the section of the *Regulatory, Safety, and Environmental Notices* that applies to your country or region.

To access this guide:

- Select the **Search** icon in the taskbar, type `HP Documentation` in the search box, and then select **HP Documentation**.

Keyboard area

Keyboards can vary by language.

NOTE: The keyboard, including the function keys and power key, is disabled in stand, tent, and tablet modes. To enable the keyboard, including the power key, change to the clamshell mode.

Touchpad

The touchpad settings and components are described here.

Touchpad settings

You learn how to adjust the touchpad settings and components here.

Adjusting touchpad settings

Use these steps to adjust touchpad settings and gestures.

1. Select the **Search** icon in the taskbar, type `touchpad settings` in the search box, and then press enter.
2. Choose a setting.

Turning on the touchpad

Follow these steps to turn on the touchpad.

1. Select the **Search** icon in the taskbar, type `touchpad settings` in the search box, and then press enter.
2. Using an external mouse, click the **Touchpad** button.

   If you are not using an external mouse, press the Tab key repeatedly until the pointer rests on the **touchpad** button. Then press the spacebar to select the button.

Touchpad components

Use the illustration and table to identify the touchpad components.

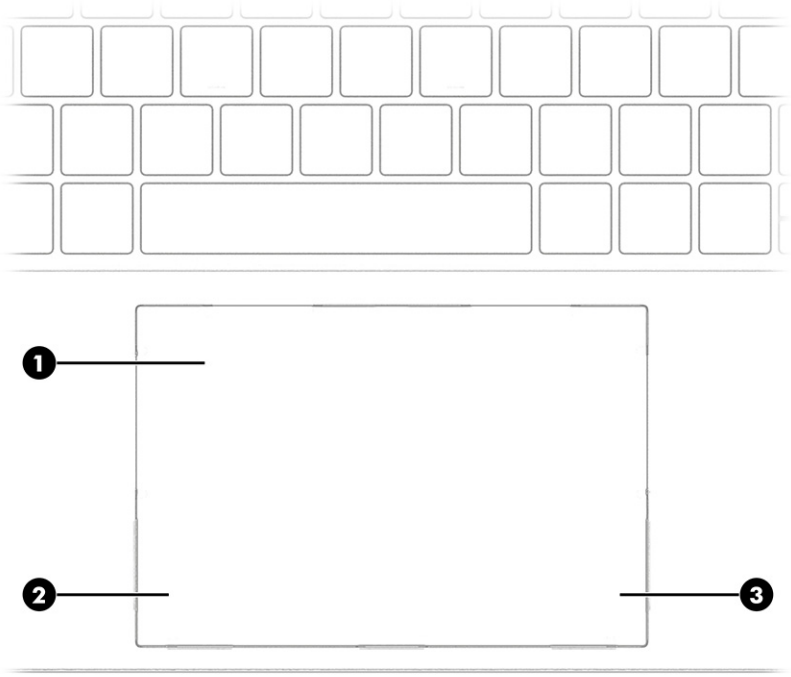


Table 2-4 Touchpad components and their descriptions

Component		Description
(1)	Touchpad zone	Reads your finger gestures to move the pointer or activate items on the screen.
(2)	Left touchpad button	Functions like the left button on an external mouse.
(3)	Right touchpad button	Functions like the right button on an external mouse.

Lights

Use the illustration and table to identify the lights on the computer.

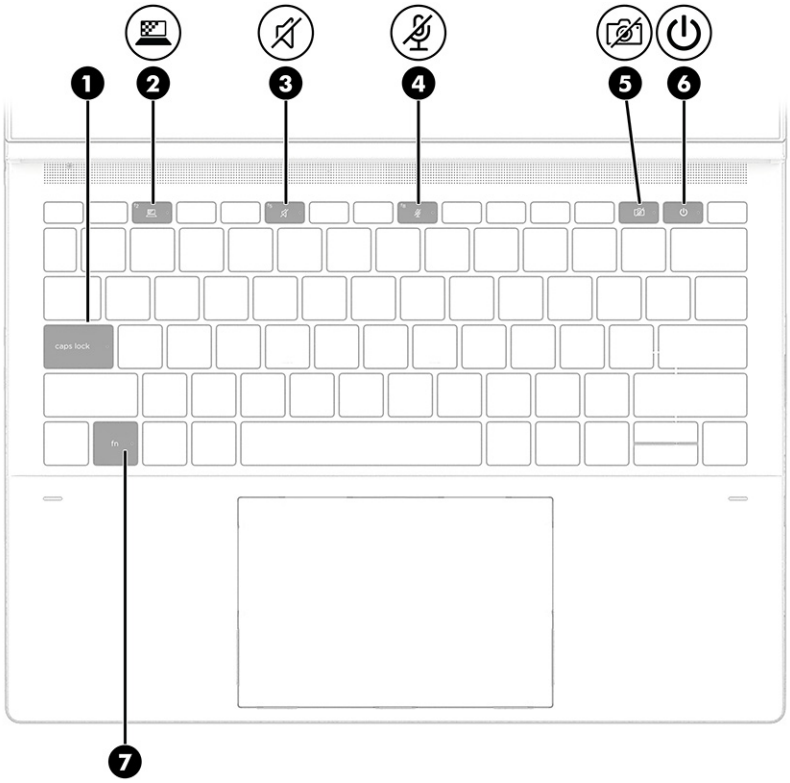


Table 2-5 Lights and their descriptions

Component		Description
(1)	Caps lock light	On: Caps lock is on, which switches the key input to all capital letters.
(2)	Privacy key light	On: Privacy screen is on, which helps prevent side-angle viewing.
(3)	Mute light	• On: Computer sound is off. • Off: Computer sound is on.
(4)	Microphone mute light	• On: Microphone is off. • Off: Microphone is on.
(5)	Camera key light	• On: The camera is off. • Off: The camera is on.
(6)	Power light	• On: The computer is on. • Blinking (select products only): The computer is in the Sleep state, a power-saving state. The computer shuts off power to the display and other unnecessary components. • Off: Depending on your computer model, the computer is off, in Hibernation, or in Sleep. Hibernation is the power-saving state that uses the least amount of power.
(7)	Fn lock light	On: The fn key is locked.

Special keys

Use the illustration and table to identify the special keys.

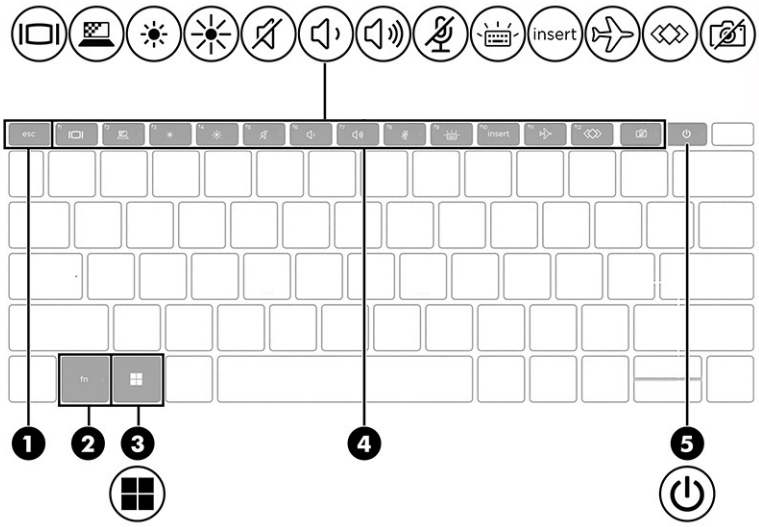


Table 2-6 Special keys and their descriptions

Component		Description
(1)	esc key	Displays system information when pressed in combination with the fn key.
(2)	fn key	Executes frequently used system functions when pressed in combination with another key. Such key combinations are called *hot keys*.
(3)	Windows key	Opens the Start menu. **NOTE:** Pressing the Windows key again will close the Start menu.
(4)	Action keys	Execute frequently used system functions.

Table 2-6 **Special keys and their descriptions (continued)**

Component		Description
(5)	Power key	• When the computer is off, press the key briefly to turn on the computer. • When the computer is on, press the key briefly to initiate Sleep. • When the computer is in the Sleep state, press the key briefly to exit Sleep (select products only). • When the computer is in Hibernation, press the key briefly to exit Hibernation. **IMPORTANT:** Pressing and holding down the power key results in the loss of unsaved information. If the computer has stopped responding and shutdown procedures are ineffective, press and hold the power key for at least 4 seconds to turn off the computer. To learn more about your power settings, use the Power icon. • Right-click the **Power** icon, and then select **Power and sleep settings**.

Bottom

Use the illustration and table to identify the bottom components.

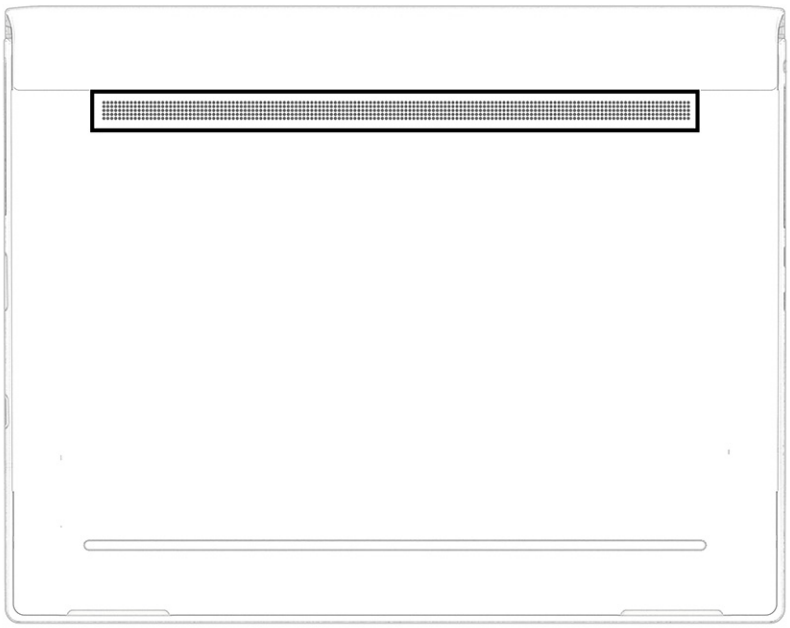


Table 2-7 **Bottom components and their descriptions**

Component	Description
Vent	Enable airflow to cool internal components. **NOTE:** The computer fan starts up automatically to cool internal components and prevent overheating. It is normal for the internal fan to cycle on and off during routine operation.

Labels

The labels affixed to the computer provide information you might need when you troubleshoot system problems or travel internationally with the computer. Labels might be in paper form or imprinted on the product.

IMPORTANT: Check the following locations for the labels described in this section: the bottom of the computer, inside the battery bay, under the service door, on the back of the display, or on the bottom of a tablet kickstand.

- Service label—Provides important information to identify your computer. When contacting support, you might be asked for the serial number, the product number, or the model number. Locate this information before you contact support.

  Your service label will resemble one of the examples shown below. Refer to the illustration that most closely matches the service label on your computer.

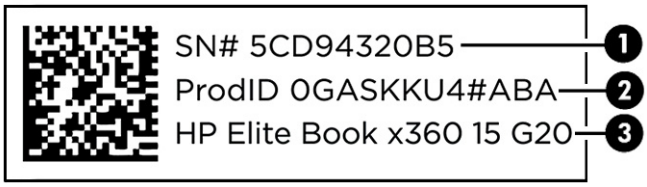


Table 2-8 **Service label components**

Component	
(1)	Serial number
(2)	Product ID
(3)	HP product name

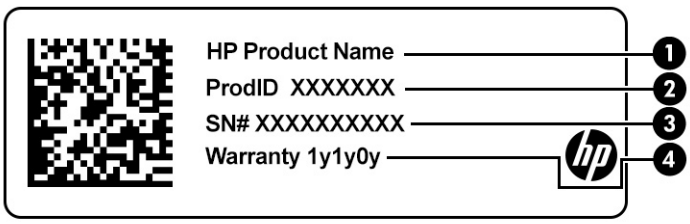


Table 2-9 **Service label components**

Component	
(1)	HP product name
(2)	Product ID
(3)	Serial number
(4)	Warranty period

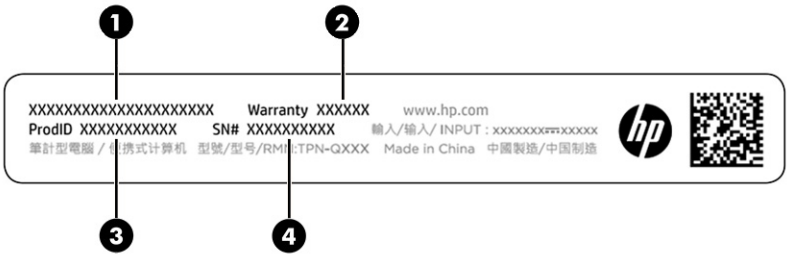


Table 2-10 **Service label components**

Component	
(1)	HP product name
(2)	Warranty period
(3)	Product ID
(4)	Serial number

- Regulatory labels—Provide regulatory information about the computer.
- Wireless certification labels—Provide information about optional wireless devices and the approval markings for the countries or regions in which the devices have been approved for use.

Using a SIM card (select products only)

Use these instructions to insert a SIM card.

IMPORTANT: You can damage the SIM card if you insert the wrong size card or insert it or the SIM card tray in the wrong direction. The card might also become stuck in the slot. Do not use SIM card adapters. To prevent damage to the SIM card or connectors, use minimal force when inserting or removing a SIM card.

Determining the correct SIM card size for your computer

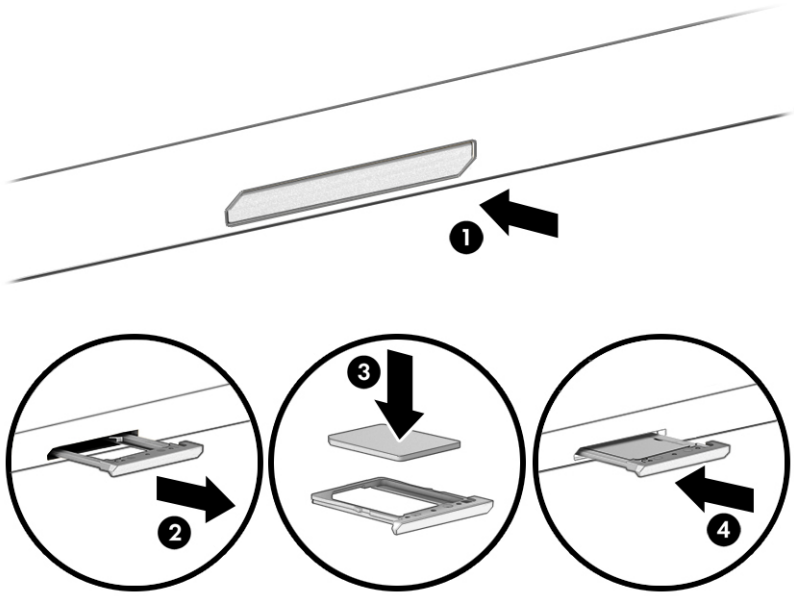
Before purchasing a SIM card, follow these instructions to determine the correct SIM card size for your computer.

1. Go to http://www.hp.com/support, and then search for your computer by product name or number.
2. Select **Product Information**.
3. Refer to the listed options to determine which card to purchase.

Inserting a nano SIM card

To insert a nano SIM card, follow these steps.

1. Turn off the computer by using the **Shut down** command.
2. Position the computer display-side up on a flat surface.
3. Press in gently on the SIM card access tray to disengage the SIM lock, and the tray will pop out of the slot **(1)**.
4. Remove the tray **(2)** from the computer and insert the card **(3)**.
5. Replace the tray in the computer. Press in gently on the tray **(4)** until it is firmly seated.



To remove the SIM card, press in gently on the SIM card access tray to disengage the SIM lock, and the tray will pop out of the slot. Remove the SIM card. Replace the tray in the computer and press in gently on the tray until it is firmly seated.

3 Illustrated parts catalog

Use this chapter to determine the spare parts that are available for the computer.

Computer major components

To identify the computer major components, use this illustration and table.

NOTE: HP continually improves and changes product parts. For complete and current information about supported parts for your computer, go to http://partsurfer.hp.com, select your country or region, and then follow the on-screen instructions.

NOTE: Details about your computer, including model, serial number, product key, and length of warranty, are on the service tag at the bottom of your computer.

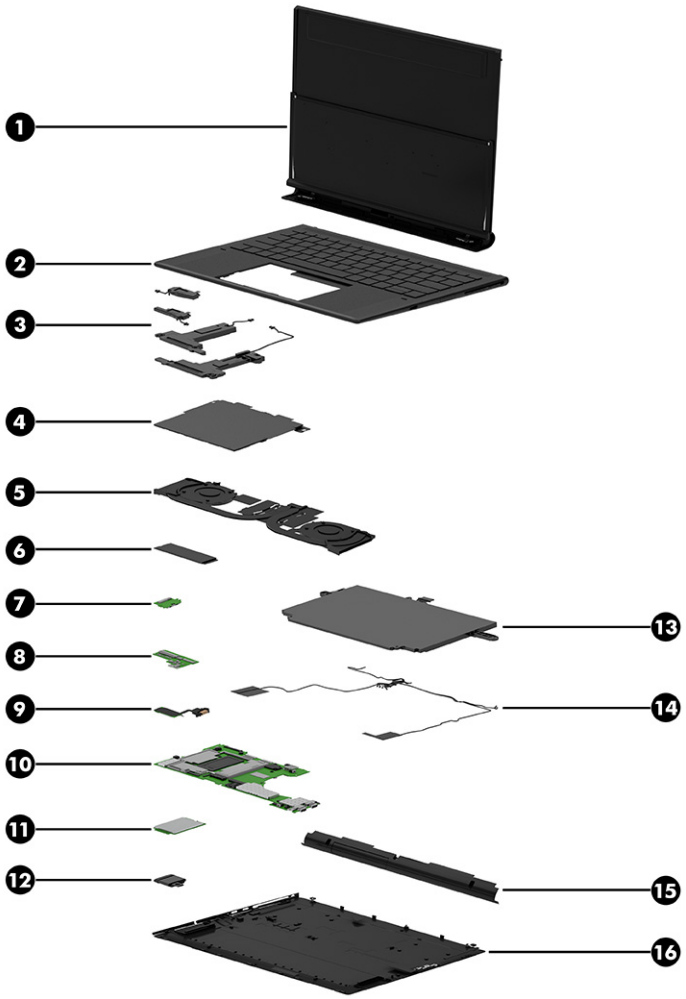


Table 3-1 Computer major component descriptions and part numbers

Item	Component	Spare part number
(1)	**Display panel** **NOTE:** Display spare parts are available only as subcomponents. For spare part information, see Display assembly subcomponents on page 18.	
(2)	**Top cover with keyboard** (includes thermal grease and replacement antennas) **NOTE:** For a detailed list of country codes, see Top cover with keyboard on page 56.	
	Privacy models with a BrightView display	N22552-xxx
	Nonprivacy models with an antiglare display	N24787-xxx
	Nonprivacy models with a BrightView display	N22553-xxx
(3)	**Speaker Kit** (includes four speakers)	N22946-001
(4)	**Touchpad** **NOTE:** The touchpad cable is available in the Cable Kit as spare part number N22929-001.	N22947-001
(5)	**Heat sink and fan assembly**	N22936-001
(6)	**Solid-state drive**	
	2 TB, PCIe-4 × 4, TLC	M52027-005
	1 TB, PCIe-4 × 4, TLC	M16560-005
	512 GB, PCIe-4 × 4, TLC	M17436-005
	512 GB, PCIe-3 × 4, TLC	L85360-001
	512 GB, PCIe-4 × 4, TLC, self-encrypting drive (SED)	M52031-005
	256 GB, PCIe-4 × 4, TLC	M52025-005
	256 GB, PCIe-4 × 4, TLC, self-encrypting drive (SED)	M52029-005
(7)	**SIM board** **NOTE:** The SIM board cable is available in the Cable Kit as spare part number N22929-001.	
	For use in models with WWAN	N22940-001
	For use in models with WLAN	N22941-001
(8)	**Transfer board** **NOTE:** The transfer board cable is available in the Cable Kit as spare part number N22929-001.	N22942-001
(9)	**Wireless charging module** **NOTE:** The wireless charging module cable is available in the Cable Kit as spare part number N22929-001.	N26149-001
(10)	**System board** All system boards use the following part numbers: xxxxxx-001: Non-Windows operating systems xxxxxx-601: Windows operating system	
	Models with integrated vPro WLAN:	
	• Intel Core i7-1265U processor and 32 GB of system memory	N20728-xxx

Table 3-1 Computer major component descriptions and part numbers (continued)

Item	Component	Spare part number
	• Intel Core i7-1265U processor and 16 GB of system memory	N20727-xxx
	• Intel Core i7-1265U processor and 16 GB of system memory (OSR)	N20726-xxx
	• Intel Core i5-1245U processor and 16 GB of system memory	N20723-xxx
	• Intel Core i5-1245U processor and 16 GB of system memory (OSR)	N20722-xxx
	Models with integrated non-vPro WLAN:	
	• Intel Core i7-1255U processor and 32 GB of system memory	N20725-xxx
	• Intel Core i7-1255U processor and 16 GB of system memory	N20724-xxx
	• Intel Core i5-1235U processor and 16 GB of system memory	N20720-xxx
	• Intel Core i5-1235U processor and 8 GB of system memory	N20721-xxx
(11)	**WWAN module**	
	Intel 5000 5G Solution WWAN	N00181-005
	Intel XMM 7560 R + LTE-Advanced Pro WWAN (Cat 16)	N00182-005
(12)	**Magnetic pen slot**	N26148-001
(13)	**Battery** (4 cell, 53 Whr, includes insulators)	M90788-005
(14)	**WWAN antenna**	
	Aux2 (4G/5G)	N22921-001
	Aux3 (4G/5G)	N22922-001
(15)	**Antenna barrel**	
	WWAN	N22928-001
	WLAN	N22927-001
(16)	**Bottom cover**	
	WWAN models	N22918-001
	WLAN models	N22919-001
*	**Audio jack**	N31280-001
*	**End covers (left and right)**	N26146-001
*	**Pen**	N02954-001
*not illustrated		

Display assembly subcomponents

To identify the display assembly subcomponents, use this illustration and table.

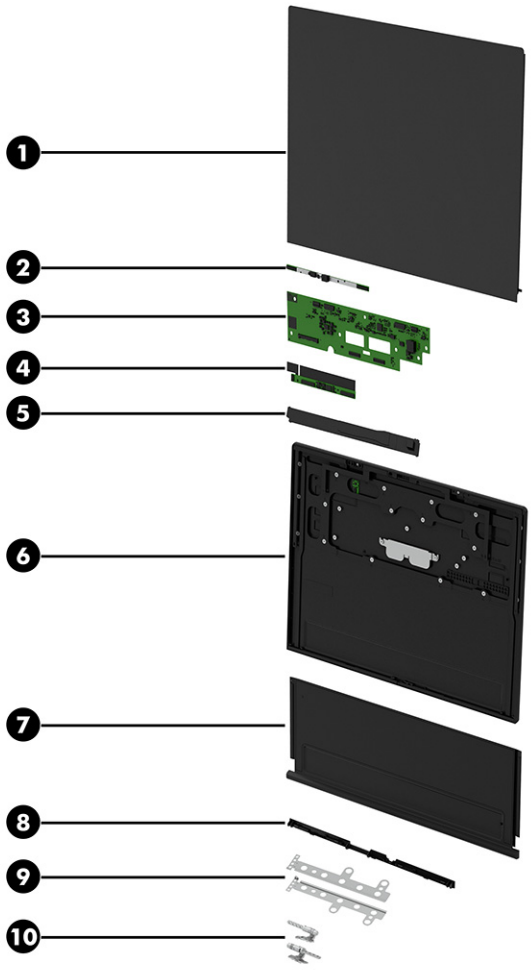


Table 3-2 **Display component descriptions and part numbers**

Item	Component	Spare part number
(1)	**Display panel**	
	WUXGA, antiglare	N23302-001
	WUXGA, BrightView	N23304-001
	WUXGA, privacy	N23303-001
	OLED	N23305-001
(2)	**Camera module**	N23306-001
(3)	**Hub board**	
	FHD models	N22937-001
	OLED models	N22938-001
(4)	**Touch control board**	N22944-001
(5)	**Display cable** (included in Display Cable Kit)	N22930-001
(6)	**LCD mid frame**	

Table 3-2 Display component descriptions and part numbers (continued)

Item	Component	Spare part number
	Models with an antiglare display	N22923-001
	Models with a BrightView display	N22924-001
(7)	**Metal back cover for hinge arm**	N22925-001
(8)	**Hinge arm cover**	N22926-001
(9)	**Hinge arm support bracket** (available in the Bracket Kit)	N22931-001
(10)	**Hinge Kit**	
	FHD displays	N22935-001
	OLED displays	N26147-001
*	**Display back cover** (metal, includes leather cover)	N22920-001
*	**Display Misc Parts Kit** (includes hinge Mylar, display insulator, display TOF rubber, bezel tape, panel tape, panel absorber, touch control board adhesive, and display gaskets)	N22934-001
*	**LCD Adhesives Kit**	N31257-001
*not illustrated		

Miscellaneous parts

To identify the miscellaneous parts, use this table.

Table 3-3 Miscellaneous part descriptions and part numbers

Component	Spare part number
AC adapter	
280 W, PFC, slim, 7.4 mm	M52952-001
120 W, PFC, 7.4 mm	L89695-001
120 W, PFC, 4.5 mm	M95377-001
65 W, nPFC, USB-C®, 1.8 m [6.0 ft], slim	L04650-850
65 W, nPFC, travel adapter	L21487-001
65 W, nPFC, USB Type-C adapter, 1.8 m (6 ft)	L67440-001
Screw Kit	N22945-001
Cable Kit (includes pen charging cable, touchpad cable, SIM cable, audio cable, and transfer board cable)	N22929-001
Display Cable Kit (includes FHD panel cable, OLED panel cable, privacy panel cable, EDP cable, camera + sensor cable, ALD cable, left IR camera cable, right RGB camera cable, TOF + microphone cable, CLS + microphone cable, touch control board cable, and eCompass board cable)	N22930-001
Bracket Kit (includes left and right hinge arm brackets, solid-state drive bracket, display bracket, PBC bracket, USB bracket, and WWAN 4G module bracket)	N22931-001
Rubber Kit (includes top bumper, WWAN module rubber, and middle and back trunk cover bumpers)	N22933-001

Table 3-3 **Miscellaneous part descriptions and part numbers (continued)**

Component	Spare part number
Miscellaneous Parts Kit (includes WWAN thermal pad, SIM tray, SIM insert, wireless charge module shield gasket, clickpad screw cover insulator, fan screw insulators, SSD Mylar shield, WWAN/SSD module Mylar absorber, 30.5 mm × 2.3 mm × 1.5 mm gasket, 40.8 mm × 2.3 mm × 2.5 mm gasket, 20 mm × 15 mm × 0.1 mm gasket, 20 mm × 13 mm ×0.8 mm gasket, 48 mm × 3 mm × 2 mm gasket, charger gasket, hinge gasket, 20 mm × 10 mm × 2.9 mm gasket, top cable insulator)	N22932-001
Wireless Bluetooth earbuds	M48196-001
USB-C-to-DisplayPort adapter	831753-001
HP USB-C-to-USB 3.0 adapter	814618-001
HP USB-C-to-VGA adapter	831751-001
USB-C to USB-A adapter	L65254-001
USB-C-to-RJ-45 adapter	918779-001
USB-C-to-HDMI™ 2.0 adapter	935325-001
USB-C male-to-USB-C male cable	L65253-001
HP Thunderbolt 120 W dock with cable	L15809-001
HP Thunderbolt 120 W G4 dock with cable	M97105-001
HP Thunderbolt 120 W G4 dock cable	M88058-001
HP Thunderbolt 280 W G4 dock with combo cable	M97106-001
HP Thunderbolt 280 W G4 dock cable	M88059-001
HP USB-C dock with cable	L64086-001
Bottom case for HP USB-C dock	L65256-001
HP USB-C/A universal dock with cable	L64087-001
Bottom case for USB-C/A universal dock	L64088-001
Universal USB-C multiport hub	M96882-001
HP 14.1 business slim top load case	L05333-001
HP Business 14.1 laptop bag	M55007-001
HP Business 17.3 laptop backpack	M55004-001
Power cord (C5, premium, 1.0 m [3.3 ft])	
Argentina	L30811-001
Australia	L22327-001
Brazil	L30812-001
Denmark	L22322-001
Europe (Austria, Belgium, Finland, France, Germany, the Netherlands, Norway, and Sweden)	L22321-001
Europe (Austria, Belgium, Finland, France, Germany, the Netherlands, Norway, and Sweden) (bundle)	N16170-001
India	L22624-001
Israel	L22323-001
Italy	L30813-001

Table 3-3 **Miscellaneous part descriptions and part numbers (continued)**

Component	Spare part number
Japan	L22330-001
North America	L22319-001
The People's Republic of China	L21930-001
South Africa	L22325-001
South Korea	L22328-001
Switzerland	L22324-001
Taiwan	L22329-001
Thailand	L22326-001
Thailand (bundle, conventional)	M85418-001
Thailand (bundle)	M85421-001
The United Kingdom	L22320-001
Power cord (C13, premium, 1.0 m [3.3 ft])	
Brazil	L32030-001
Italy	L32031-001
Duckhead power cord	
Japan	L33157-001
Power cords (C5, conventional, 1.0 m [3.3 ft])	
Argentina	L19357-001
Australia	L19358-001
Denmark	L19360-001
Europe (Austria, Belgium, Finland, France, Germany, the Netherlands, Norway, and Sweden)	L19361-001
India	L19363-001
Israel	L19362-001
Italy	L19364-001
Japan	L19365-001
North America	L19367-001
People's Republic of China	L19368-001
South Africa	L19369-001
South Korea	L19366-001
Switzerland	L19370-001
Taiwan	L19372-001
Thailand	M85418-001
United Kingdom	L19373-001
Power cords (C5, conventional, 1.8 m [6.0 ft])	

Table 3-3 **Miscellaneous part descriptions and part numbers (continued)**

Component	Spare part number
Argentina	L19357-002
Australia	L19358-002
Denmark	L19360-002
Europe (Austria, Belgium, Finland, France, Germany, the Netherlands, Norway, and Sweden)	L19361-002
India	L19363-002
Israel	L19362-002
Italy	L19364-002
Japan	L19365-002
North America	L19367-002
People's Republic of China	L19368-002
South Africa	L19369-002
South Korea	L19366-002
Switzerland	L19370-002
Taiwan	L19372-002
Thailand	M85418-002
United Kingdom	L19373-001
Power cords (C5, conventional, right angle, 1.0 m [3.3 ft])	
Argentina	931249-001
Brazil	923430-002
Denmark	923430-003
Europe (Austria, Belgium, Finland, France, Germany, the Netherlands, Norway, and Sweden)	923430-004
Italy	923430-007
North America	923430-008
Power cords (C13, premium, 1.0 m [3.3 ft])	
Australia	100661-021
Denmark	130627-014
Europe (Austria, Belgium, Finland, France, Germany, the Netherlands, Norway, and Sweden)	100614-016
India	403440-008
Israel	398062-011
Italy	109197-012
Japan	653326-005
North America	121565-023
People's Republic of China	286496-024
South Africa	187487-012

Table 3-3 Miscellaneous part descriptions and part numbers (continued)

Component	Spare part number
South Korea	231216-015
Switzerland	150304-015
Taiwan	393312-008
Thailand	285052-013
United Kingdom	100613-021

4 Removal and replacement procedures preliminary requirements

Use this information to properly prepare to disassemble and reassemble the computer.

Tools required

You need the following tools to complete the removal and replacement procedures:

- Tweezers
- Nonconductive, nonmarking pry tool
- Magnetic Phillips P0 screwdriver
- Magnetic Phillips P1 screwdriver

Service considerations

The following sections include some of the considerations that you must keep in mind during disassembly and assembly procedures.

NOTE: As you remove each subassembly from the computer, place the subassembly (and all accompanying screws) away from the work area to prevent damage.

Plastic parts

Using excessive force during disassembly and reassembly can damage plastic parts.

Cables and connectors

Handle cables with extreme care to avoid damage.

IMPORTANT: When servicing the computer, be sure that cables are placed in their proper locations during the reassembly process. Improper cable placement can damage the computer.

Apply only the tension required to unseat or seat the cables during removal and insertion. Handle cables by the connector whenever possible. In all cases, avoid bending, twisting, or tearing cables. Be sure that cables are routed so that they cannot be caught or snagged as you remove or replace parts. Handle flex cables with extreme care; these cables tear easily.

Drive handling

Note the following guidelines when handling drives.

IMPORTANT: Drives are fragile components. Handle them with care. To prevent damage to the computer, damage to a drive, or loss of information, observe these precautions:

- Before removing or inserting a hard drive, shut down the computer. If you are unsure whether the computer is off or in Hibernation, turn the computer on, and then shut it down through the operating system.
- Before handling a drive, be sure that you are discharged of static electricity. While handling a drive, avoid touching the connector.
- Before removing an optical drive, be sure that a disc is not in the drive, and be sure that the optical drive tray is closed.
- Handle drives on surfaces covered with at least 2.54 cm (1 inch) of shock-proof foam.
- Avoid dropping drives from any height onto any surface.
- After removing a hard drive or an optical drive, place it in a static-proof bag.
- Avoid exposing an internal hard drive to products that have magnetic fields, such as monitors or speakers.
- Avoid exposing a drive to temperature extremes or liquids.
- If a drive must be mailed, place the drive in a bubble pack mailer or other suitable form of protective packaging, and label the package “FRAGILE.”

Electrostatic discharge information

A sudden discharge of static electricity from your finger or other conductor can destroy static-sensitive devices or microcircuitry. Often the spark is neither felt nor heard, but damage occurs. An electronic device exposed to electrostatic discharge (ESD) might not appear to be affected at all and can work perfectly throughout a normal cycle. The device might function normally for a while, but it has been degraded in the internal layers, reducing its life expectancy.

Networks built into many integrated circuits provide some protection, but in many cases, the discharge contains enough power to alter device parameters or melt silicon junctions.

IMPORTANT: To prevent damage to the device when you remove or install internal components, observe these precautions:

- Keep components in their electrostatic-safe containers until you are ready to install them.
- Before touching an electronic component, discharge static electricity by using the guidelines described in Personal grounding methods and equipment on page 27.
- Avoid touching pins, leads, and circuitry. Handle electronic components as little as possible.
- If you remove a component, place it in an electrostatic-safe container.

Generating static electricity

Follow these static electricity guidelines:

- Different activities generate different amounts of static electricity.
- Static electricity increases as humidity decreases.

Table 4-1 **Static electricity occurrence based on activity and humidity**

	Relative humidity		
Event	**55%**	**40%**	**10%**
Walking across carpet	7,500 V	15,000 V	35,000 V
Walking across vinyl floor	3,000 V	5,000 V	12,000 V
Motions of bench worker	400 V	800 V	6,000 V
Removing dual in-line packages (DIPs) from plastic tube	400 V	700 V	2,000 V
Removing DIPs from vinyl tray	2,000 V	4,000 V	11,500 V
Removing DIPs from polystyrene foam	3,500 V	5,000 V	14,500 V
Removing bubble pack from PCB (printed circuit board)	7,000 V	20,000 V	26,500 V
Packing PCBs in foam-lined box	5,000 V	11,000 V	21,000 V
Multiple electric components can be packaged together in plastic tubes, trays, or polystyrene foam.			

NOTE: As little as 700 V of static electricity can degrade a product.

Preventing electrostatic damage to equipment

Many electronic components are sensitive to ESD. Circuitry design and structure determine the degree of sensitivity.

The following packaging and grounding precautions are necessary to prevent static electricity damage to electronic components:

- To avoid hand contact, transport products in static-safe containers such as tubes, bags, or boxes.
- Protect all electrostatic parts and assemblies with conductive or approved containers or packaging.
- Keep electrostatic-sensitive parts in their containers until they arrive at static-free stations.
- Place items on a grounded surface before removing them from their container.
- Always be properly grounded when touching a sensitive component or assembly.
- Avoid contact with pins, leads, or circuitry.
- Place reusable electrostatic-sensitive parts from assemblies in protective packaging or conductive foam.

Personal grounding methods and equipment

Using certain equipment can prevent static electricity damage to electronic components.

- **Wrist straps** are flexible straps with a maximum of 1 MΩ ±10% resistance in the ground cords. To provide proper ground, a strap must be worn snug against bare skin. The ground cord must be connected and fit snugly into the banana plug connector on the grounding mat or workstation.
- You can use **heel straps, toe straps, and boot straps** at standing workstations. These straps are compatible with most types of shoes or boots. On conductive floors or dissipative floor mats, use them on both feet with a maximum of 1 MΩ ±10% resistance between the operator and ground.

Table 4-2 **Static shielding protection levels**

Static shielding protection levels	
Method	**Voltage**
Antistatic plastic	1,500
Carbon-loaded plastic	7,500
Metallized laminate	15,000

Grounding the work area

To prevent static damage at the work area, follow these precautions:

- Cover the work surface with approved static-dissipative material.
- Use a wrist strap connected to a properly grounded work surface and use properly grounded tools and equipment.
- Use static-dissipative mats, foot straps, or air ionizers to give added protection.
- Handle electrostatic sensitive components, parts, and assemblies by the case or PCB laminate. Handle them only at static-free work areas.
- Turn off power and input signals before inserting and removing connectors or test equipment.
- Use fixtures made of static-safe materials when fixtures must directly contact dissipative surfaces.
- Keep the work area free of nonconductive materials, such as ordinary plastic assembly aids and polystyrene foam.
- Use conductive field service tools, such as cutters, screwdrivers, and vacuums.
- Avoid contact with pins, leads, or circuitry.

Recommended materials and equipment

HP recommends certain materials and equipment to prevent static electricity:

- Antistatic tape
- Antistatic smocks, aprons, or sleeve protectors
- Conductive bins and other assembly or soldering aids
- Conductive foam
- Conductive tabletop workstations with ground cord of 1 MΩ ±10% resistance
- Static-dissipative table or floor mats with hard tie to ground
- Field service kits
- Static awareness labels
- Wrist straps and footwear straps providing 1 MΩ ±10% resistance
- Material handling packages

- Conductive plastic bags
- Conductive plastic tubes
- Conductive tote boxes
- Opaque shielding bags
- Transparent metallized shielding bags
- Transparent shielding tubes

Cleaning your computer

Cleaning your computer regularly removes dirt and debris so that your device continues to operate at its best. Use the following information to safely clean the external surfaces of your computer.

Enabling HP Easy Clean (select products only)

HP Easy Clean helps you to avoid accidental input while you clean the computer surfaces. This software disables devices such as the keyboard, touch screen, and touchpad for a preset amount of time so that you can clean all computer surfaces.

1. Start HP Easy Clean in one of the following ways:
   - Select the **Start** menu, and then select **HP Easy Clean**.

     – or –
   - Select the **HP Easy Clean** icon in the taskbar.

     – or –
   - Select **Start**, and then select the **HP Easy Clean** tile.
2. Now that your device is disabled for a short period, see Removing dirt and debris from your computer on page 29 for the recommended steps to clean the high-touch, external surfaces on your computer. After you remove the dirt and debris, you can also clean the surfaces with a disinfectant.

Removing dirt and debris from your computer

Here are the recommended steps to clean dirt and debris from your computer.

1. Wear disposable gloves made of latex (or nitrile gloves, if you are latex-sensitive) when cleaning the surfaces.
2. Turn off your device and unplug the power cord and other connected external devices. Remove any installed batteries from items such as wireless keyboards.

⚠ **CAUTION:** To prevent electric shock or damage to components, never clean a product while it is turned on or plugged in.

3. Moisten a microfiber cloth with water. The cloth should be moist, but not dripping wet.

IMPORTANT: To avoid damaging the surface, avoid abrasive cloths, towels, and paper towels.

4. Wipe the exterior of the product gently with the moistened cloth.

IMPORTANT: Keep liquids away from the product. Avoid getting moisture in any openings. If liquid makes its way inside your HP product, it can cause damage to the product. Do not spray liquids directly on the product. Do not use aerosol sprays, solvents, abrasives, or cleaners containing hydrogen peroxide or bleach that might damage the finish.

5. Start with the display (if applicable). Wipe carefully in one direction, and move from the top of the display to the bottom. Finish with any flexible cables, like power cord, keyboard cable, and USB cables.
6. Be sure that surfaces have completely air-dried before turning the device on after cleaning.
7. Discard the gloves after each cleaning. Clean your hands immediately after you remove the gloves.

Packaging and transporting guidelines

Follow these grounding guidelines when packaging and transporting equipment:

- To avoid hand contact, transport products in static-safe tubes, bags, or boxes.
- Protect ESD-sensitive parts and assemblies with conductive or approved containers or packaging.
- Keep ESD-sensitive parts in their containers until the parts arrive at static-free workstations.
- Place items on a grounded surface before removing items from their containers.
- Always be properly grounded when touching a component or assembly.
- Store reusable ESD-sensitive parts from assemblies in protective packaging or nonconductive foam.
- Use transporters and conveyors made of antistatic belts and roller bushings. Be sure that mechanized equipment used for moving materials is wired to ground and that proper materials are selected to avoid static charging. When grounding is not possible, use an ionizer to dissipate electric charges.

Accessing support information

To find the HP support that you need, use this information.

Table 4-3 Support information locations

Service consideration	Path to access information
Records of reported failure incidents stored on the computer	**Windows:** Preoperating system failures are logged in the BIOS Event Log. To view the BIOS Event Log: 1. Press the power button. 2. Immediately and repeatedly press esc when the power button light turns white. **NOTE:** If you do not press esc at the appropriate time, you must restart the computer and again repeatedly press esc when the power button light turns white to access the utility. 3. Press f10 to enter the BIOS setup. 4. (On commercial products) Under the **Main** tab, select **BIOS event log**, and then select **View BIOS Event Log**. - or - (On consumer products) Under the **Main** tab, select **System Log**. Post-operating system failures are logged in the Event Viewer. 1. Turn on the computer and allow the operating system to open. 2. Select the search icon in the taskbar. 3. Type `Event Viewer`, and then press enter. 4. Select the log from the left panel. Details display in the right panel. **Chrome:** 1. Go to support.google.com/chrome. 2. Search `collect Chrome device logs`.
Technical bulletins	To locate technical bulletins: 1. Go to www.hp.com. 2. Place the cursor over **Problem solving** to display more options. 3. Select **Support & Troubleshooting**. 4. Type the serial number, product number, or product name to go to the product support page. 5. Select **Advisories** to view technical bulletins.
Repair professionals	To locate repair professionals: 1. Go to www.hp.com. 2. Place the cursor over **Support resources** to display more options. 3. Select **Authorized service providers**.

Table 4-3 Support information locations (continued)

Service consideration	Path to access information
Component and diagnosis information, failure detection, and required action	To locate diagnosis information and actions: 1. Go to http://www.hp.com/go/techcenter/pcdiags. 2. Select **Get Support**. 3. Near the bottom of the window, select **Notebook PCs**, and then select your location.

5 Removal and replacement procedures for authorized service provider parts

This chapter provides removal and replacement procedures for authorized service provider parts.

IMPORTANT: Only an authorized service provider should access the components described in this chapter. Accessing these parts can damage the computer or void the warranty.

NOTE: Details about your computer, including model, serial number, product key, and length of warranty, are on the service tag at the bottom of your computer.

Component replacement procedures

To remove and replace computer components, use the procedures described in this section.

NOTE: HP continually improves and changes product parts. For complete and current information about supported parts for your computer, go to http://partsurfer.hp.com, select your country or region, and then follow the on-screen instructions.

Make special note of each screw size and location during removal and replacement.

Preparation for disassembly

To prepare to disassemble the computer, use these steps.

See Removal and replacement procedures preliminary requirements on page 25 for initial safety procedures.

1. Turn off the computer. If you are unsure whether the computer is off or in Hibernation, turn the computer on, and then shut it down through the operating system.
2. Disconnect the power from the computer by unplugging the power cord from the computer.
3. Disconnect all external devices from the computer.

Bottom cover

To remove the bottom cover, use this procedure and illustration.

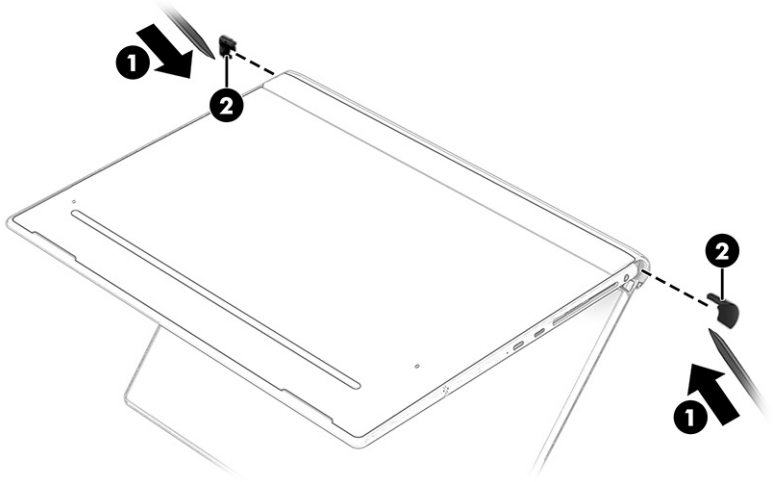
Table 5-1 Bottom cover descriptions and part numbers

Description	Spare part number
Bottom cover (WWAN models)	N22918-001
Bottom cover (WLAN models)	N22919-001
End covers (left and right)	N26146-001

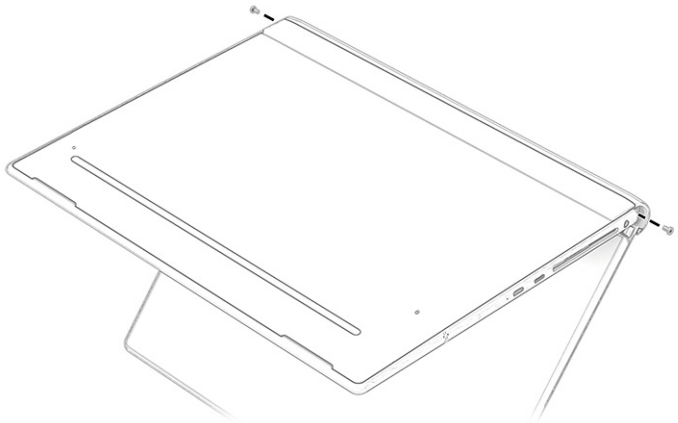
Before loosening the bottom cover, prepare the computer for disassembly (see Preparation for disassembly on page 33).

Loosen the bottom cover:

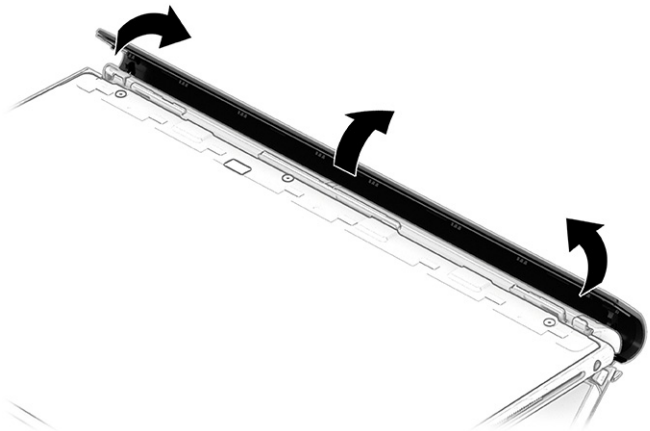
1. Open the display to approximately 110° so there is slack in the cover where it wraps over the back of the computer.
2. Use a tool **(1)** to release the cover plugs from the inside of the cover on both ends of the computer, and then remove the plugs **(2)**.



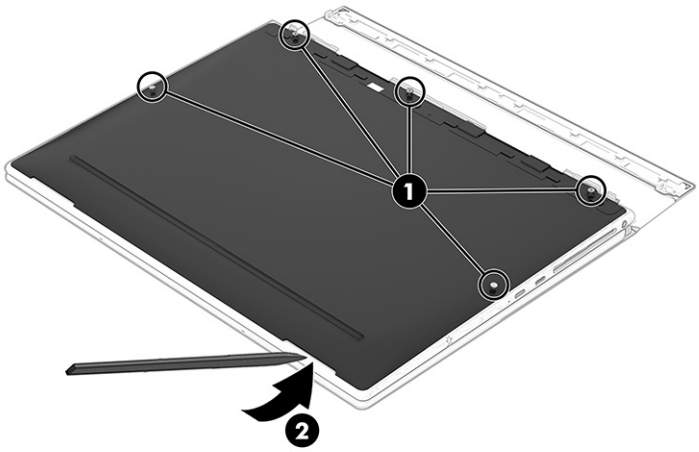
3. Remove the Phillips M2.0 × 4.0 screw from under each cover plug.



4. The leather cover extends around the back of the computer and onto the bottom cover. Push the outside edges of the leather cover at an angle from the outside toward the inside to release the sides of the cover, and then push the middle of the cover off the computer.



5. Loosen the five captive Torx T5 screws **(1)**, and then insert a tool into one of the slots on the bottom edge of the cover and lift the tool to release the cover **(2)**.



To replace the bottom cover, reverse the removal procedures.

NOTE: When replacing the bottom cover, be sure to toe the cover in at the bottom and clip it in place up the sides. Be sure all clips are reinserted before installing the screws in the sides.

Solid-state drive

To remove the M.2 solid-state drive, use this procedure and illustration.

Table 5-2 Solid-state drive descriptions and part numbers

Description	Spare part number
2 TB, PCIe-4 × 4, TLC	M52027-005
1 TB, PCIe-4 × 4, TLC	M16560-005

Table 5-2 **Solid-state drive descriptions and part numbers (continued)**

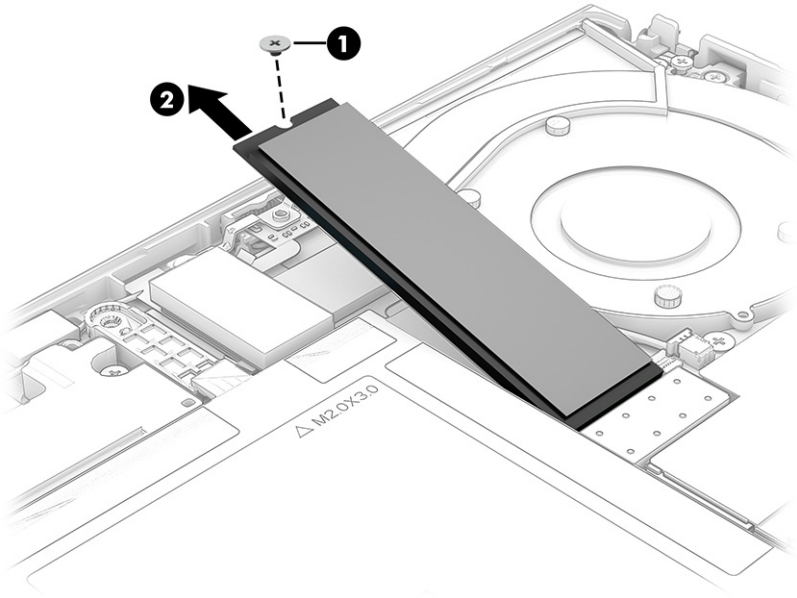
Description	Spare part number
512 GB, PCIe-4 × 4, TLC	M17436-005
512 GB, PCIe-3 × 4, TLC	L85360-001
512 GB, PCIe-4 × 4, TLC, self-encrypting drive (SED)	M52031-005
256 GB, PCIe-4 × 4, TLC	M52025-005
256 GB, PCIe-4 × 4, TLC, self-encrypting drive (SED)	M52029-005
Solid-state drive bracket (available in the Bracket Kit)	N22931-001

Before removing the solid-state drive, follow these steps:

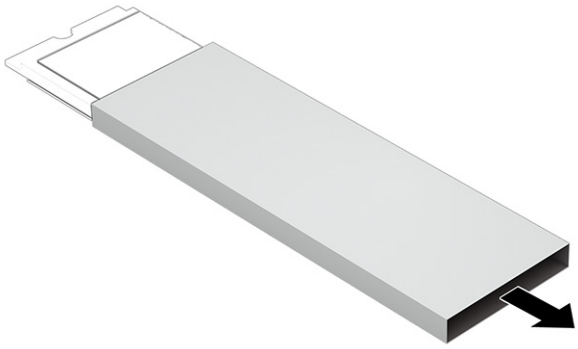
1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the bottom cover (see Bottom cover on page 33).
3. Disconnect the battery cable from the system board (see Battery on page 42).

Remove the solid-state drive:

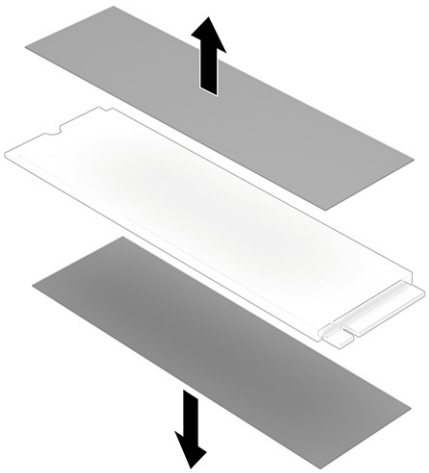
1. Remove the Phillips M1.4 × 1.2 screw **(1)** that secures the drive to the computer.
2. Pull the drive away from the socket to remove it **(2)**.



3. Slide the sleeve off the solid-state drive to remove it.



4. Remove the Mylar from both sides of the solid-state drive. Install the Mylar on a new drive.



To install the solid-state drive, reverse the removal procedures.

NOTE: Solid-state drives are designed with a notch to prevent incorrect insertion.

Magnetic pen slot

To remove the magnetic pen slot, use this procedure and illustration.

Table 5-3 Magnetic pen slot description and part number

Description	Spare part number
Magnetic pen slot	N26148-001

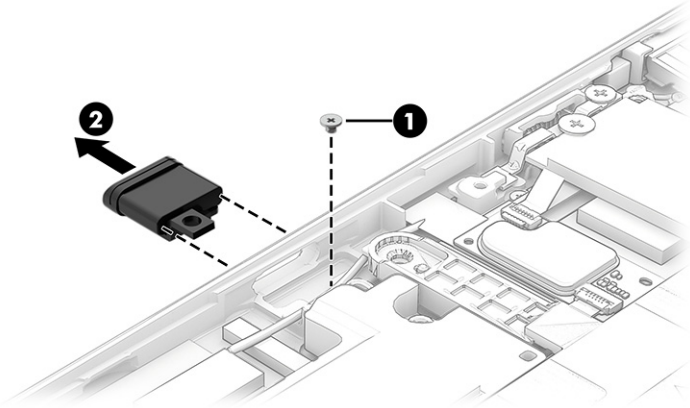
Before removing the magnetic pen slot, follow these steps:

1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the bottom cover (see Bottom cover on page 33).

3. Disconnect the battery cable from the system board (see Battery on page 42).

Remove the magnetic pen slot:

- Remove the Phillips M2.0 × 3.0 screw **(1)** that secures the magnetic pen slot to the computer, and then slide the slot out of the computer **(2)**.



To install the magnetic pen slot, reverse this procedure.

SIM board

To remove the SIM board, use this procedure and illustration.

Table 5-4 SIM board description and part number

Description	Spare part number
SIM board for use in models with WWAN	N22940-001
SIM board for use in models with WLAN	N22941-001
SIM cable (available in Cable Kit)	N22929-001
SIM tray and SIM insert (available in Misc Parts Kit)	N22932-001

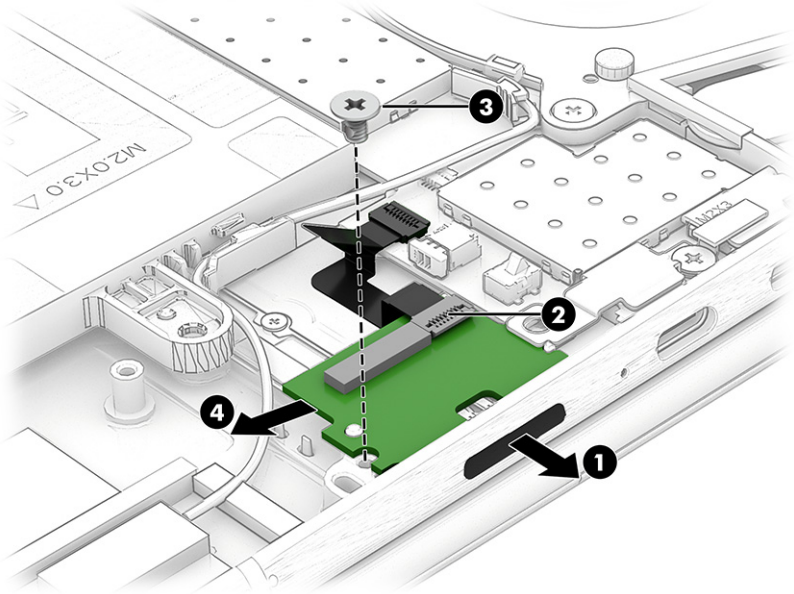
Before removing the SIM board, follow these steps:

1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the bottom cover (see Bottom cover on page 33).
3. Disconnect the battery cable from the system board (see Battery on page 42).

Remove the SIM board:

1. Remove the SIM holder from the side of the computer **(1)**.
2. Disconnect the cable from the ZIF connector on the SIM board **(2)**.
3. Remove the Phillips M2.0 × 3.0 screw **(3)** that secures the SIM board to the computer.

4. Lift the side of the board up to approximately 20°, and then pull the board out from under the USB bracket **(4)**.



To install the SIM board, reverse this procedure.

NOTE: When replacing the SIM board, be sure to insert the board under the USB bracket.

WWAN module

To remove the WWAN module, use this procedure and illustration.

Table 5-5 WWAN module descriptions and part numbers

Description	Spare part number
Intel 5000 5G Solution WWAN	N00181-005
Intel XMM 7560 R + LTE-Advanced Pro WWAN (Cat 16)	N00182-005
WWAN card rubber (included in Rubber Kit)	N22933-001
WWAN module bracket (available in the Bracket Kit)	N22931-001

IMPORTANT: To prevent an unresponsive system, replace the wireless module only with a wireless module authorized for use in the computer by the governmental agency that regulates wireless devices in your country or region. If you replace the module and then receive a warning message, remove the module to restore device functionality, and then contact technical support.

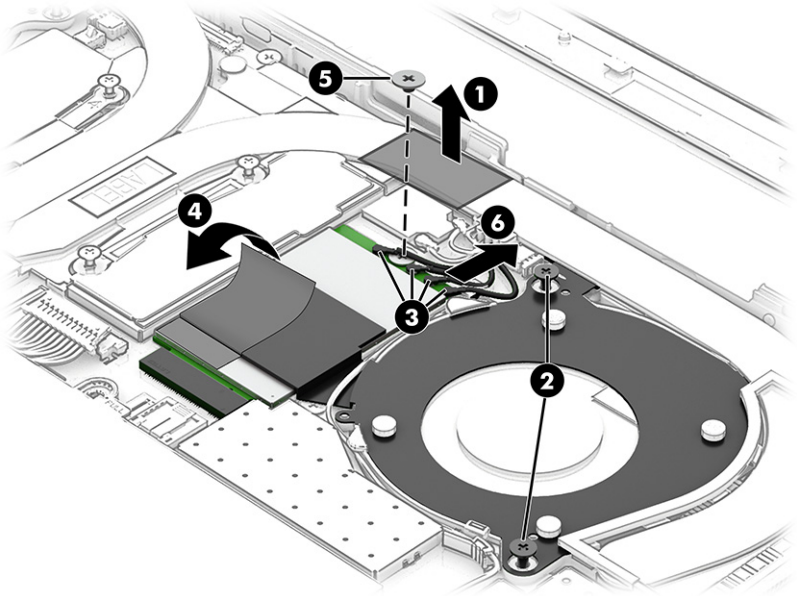
Before removing the WWAN module, follow these steps:

1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the bottom cover (see Bottom cover on page 33).
3. Disconnect the battery cable from the system board (see Battery on page 42).

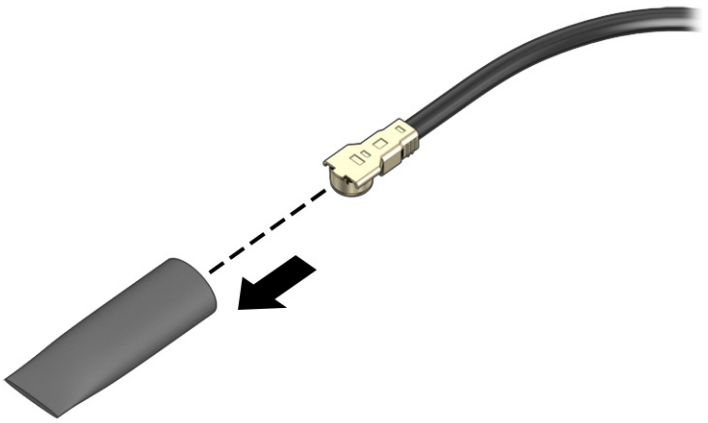
Remove the WWAN module:

1. Remove the tape from the connectors on the module **(1)**.
2. Loosen the two captive screws from the right fan **(2)**.
3. Carefully disconnect the antenna cables from the connectors on the module **(3)**.
4. Peel the gray conductive tape off the module **(4)**. You can leave the tape connected to the connector.
5. Remove the Phillips M1.6 × 1.6 screw **(5)**, and then pull the module out from under the plastic that extends over it from the thermal module **(6)**.

NOTE: 4G WWAN modules come with a reusable bracket. Be sure to use this bracket when installing a 4G WWAN module.



6. If the WWAN antenna is not connected to the terminal on the WWAN module, install a protective sleeve on the antenna connector, as shown in the following illustration.



To install the WWAN module, reverse this procedure.

NOTE: Be sure to tighten the two screws on the right fan after installing a WWAN module.

Wireless charging module

To remove the wireless charging module, use this procedure and illustration.

Table 5-6 Wireless charging module description and part number

Description	Spare part number
Wireless charging module	N26149-001
Wireless charging module cable (available in Cable Kit)	N22929-001

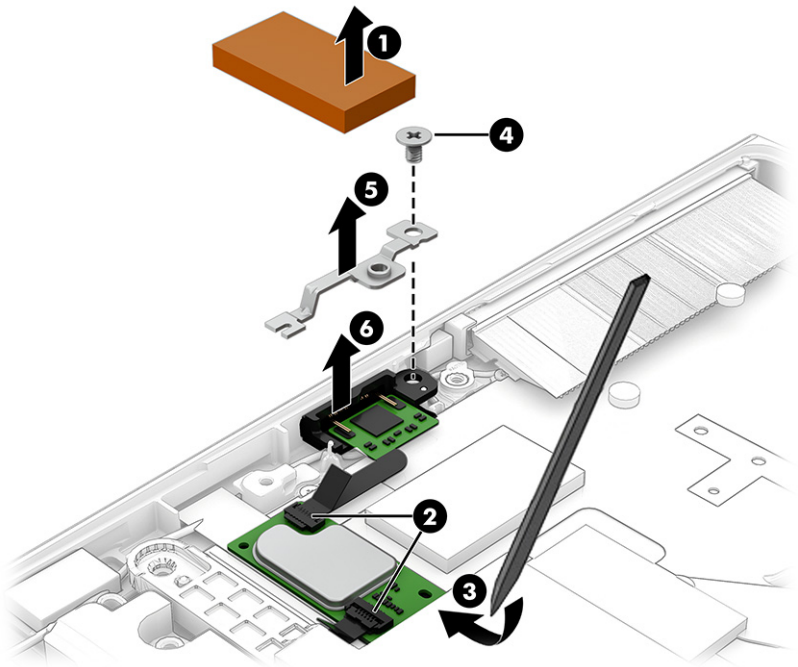
Before removing the wireless charging module, follow these steps:

1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the bottom cover (see Bottom cover on page 33).
3. Disconnect the battery cable from the system board (see Battery on page 42).

Remove the wireless charging module:

1. Remove the rubber piece from the board **(1)**.
2. Disconnect the cables from the ZIF connectors on the top and bottom of the board **(2)**.
3. Use a tool to release the board **(3)**. The board is secured with adhesive.
4. Remove the Phillips M2.0 × 3.0 screw **(4)** that secures the charger bracket to the computer, and then remove the bracket **(5)**.

5. Pull the wireless charger into the computer to remove it **(6)**.



To install the wireless charging module, reverse this procedure.

Battery

To remove the battery, use this procedure and illustration.

Table 5-7 Battery description and part number

Description	Spare part number
Battery, 4 cell, 53 Whr (includes insulators)	M90788-005

⚠ **WARNING!** To avoid personal injury and damage to the product:

- Do *not* puncture, twist, or crack the battery.
- Do *not* cause an external puncture or rupture to the battery. They can cause a short inside the battery, which can result in battery thermal runaway.
- Do *not* handle or touch the battery enclosure with sharp objects such as tweezers or pliers, which might puncture the battery.
- Do *not* compress or squeeze the battery case with tools or heavy objects stacked on top of the case. These actions can apply undue force on the battery.
- Do *not* touch the connectors with any metallic surface or object, such as metal tools, screws, or coins, which can cause shorting across the connectors.

Before removing the battery, follow these steps:

1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).

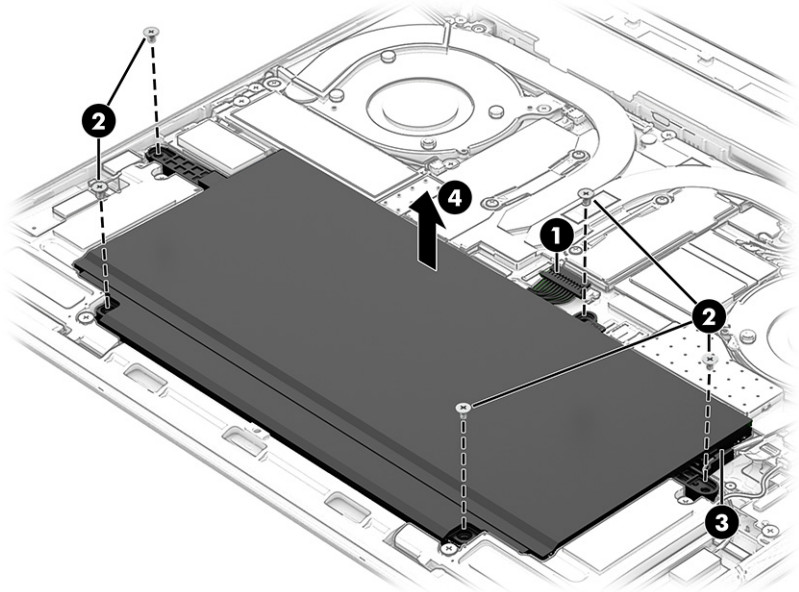
2. Remove the bottom cover (see Bottom cover on page 33).

WARNING! To reduce potential safety issues, use only the user-replaceable battery provided with the computer, a replacement battery provided by HP, or a compatible battery purchased from HP.

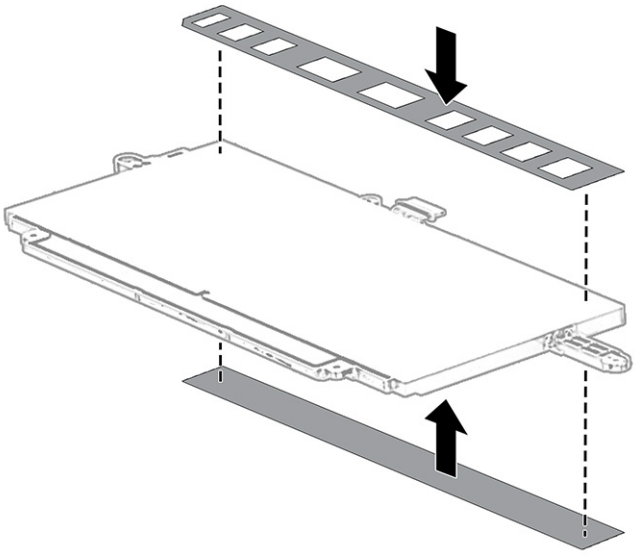
IMPORTANT: Removing a battery that is the sole power source for the computer can cause loss of information. To prevent loss of information, save your work or shut down the computer through Windows before you remove the battery.

Remove the battery:

1. Disconnect the battery cable from the system board **(1)**.
2. Remove the five Phillips M2.0 × 3.0 screws **(2)** that secure the battery to the computer.
3. Remove the antenna cable from the clip on the right side of the battery **(3)**.
4. Remove the battery from the computer **(4)**.



5. When installing a battery, be sure to install the insulators on the top and bottom of the battery as shown in the following illustration.



To install the battery, reverse the removal procedures.

Bottom speakers

To remove the bottom speakers, use this procedure and illustration.

Table 5-8 Bottom speakers description and part number

Description	Spare part number
Speakers (includes all speakers)	N22946-001

Before removing the speakers, follow these steps:

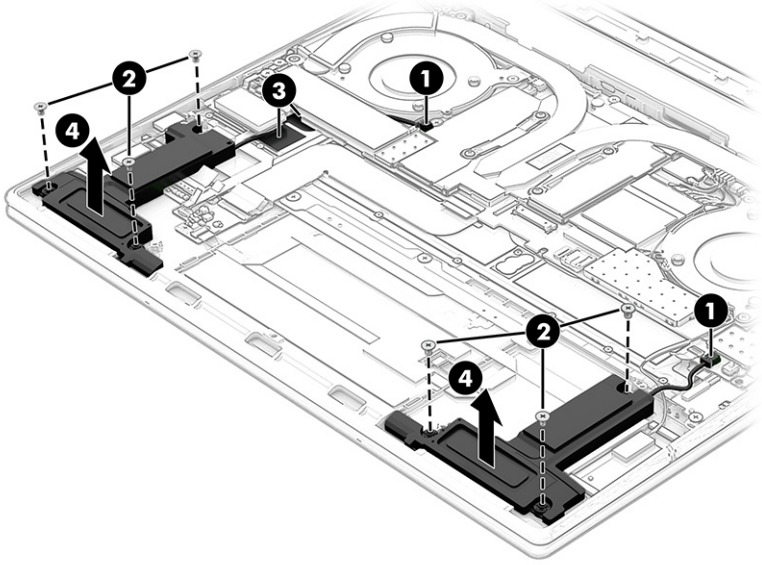
1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the bottom cover (see Bottom cover on page 33).
3. Remove the battery (see Battery on page 42).

Remove the bottom speakers:

1. Disconnect both speaker cables from the system board **(1)**.
2. Remove the three Phillips M2.0 × 3.0 screws **(2)** from each speaker.
3. Remove the speaker cable from the two Mylar tabs that secure it **(3)**.

NOTE: To avoid tearing the tabs, gently pull up on the tabs when releasing the cable.

4. Remove the speakers from the computer **(4)**.



To install the bottom speakers, reverse this procedure.

NOTE: When installing the speakers, be sure rubber washers are installed in the screw holes in the speakers.

Transfer board

To remove the transfer board, use this procedure and illustration.

Table 5-9 Transfer board description and part number

Description	Spare part number
Transfer board	N22942-001
Transfer board cable (available in Cable Kit)	N22929-001

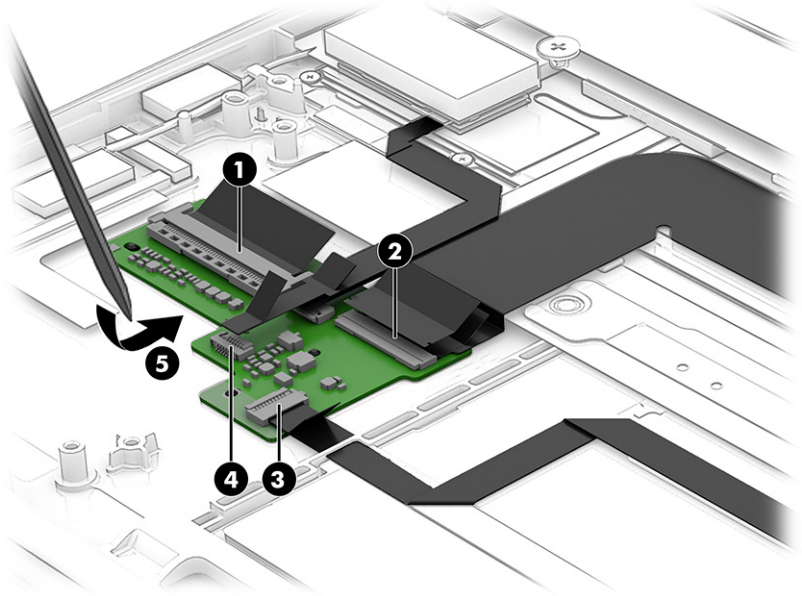
Before removing the transfer board, follow these steps:

1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the bottom cover (see Bottom cover on page 33).
3. Remove the battery (see Battery on page 42).
4. Remove the bottom speakers (see Bottom speakers on page 44).

Remove the transfer board:

1. Disconnect the keyboard cable from the transfer board **(1)**.
2. Disconnect the system board cable from the transfer board **(2)**.
3. Disconnect the touchpad cable from the transfer board **(3)**.
4. Disconnect the backlight cable from the transfer board **(4)**.

5. Use a tool to release the board from the computer **(5)**. The board is secured with adhesive.



To install the transfer board, reverse this procedure.

Touchpad

To remove the touchpad, use this procedure and illustration.

Table 5-10 Touchpad description and part number

Description	Spare part number
Touchpad	N22947-001
Touchpad cable (included in Cable Kit)	N22929-001

Before removing the touchpad, follow these steps:

1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the bottom cover (see Bottom cover on page 33).
3. Remove the battery (see Battery on page 42).
4. Remove the bottom speakers (see Bottom speakers on page 44).

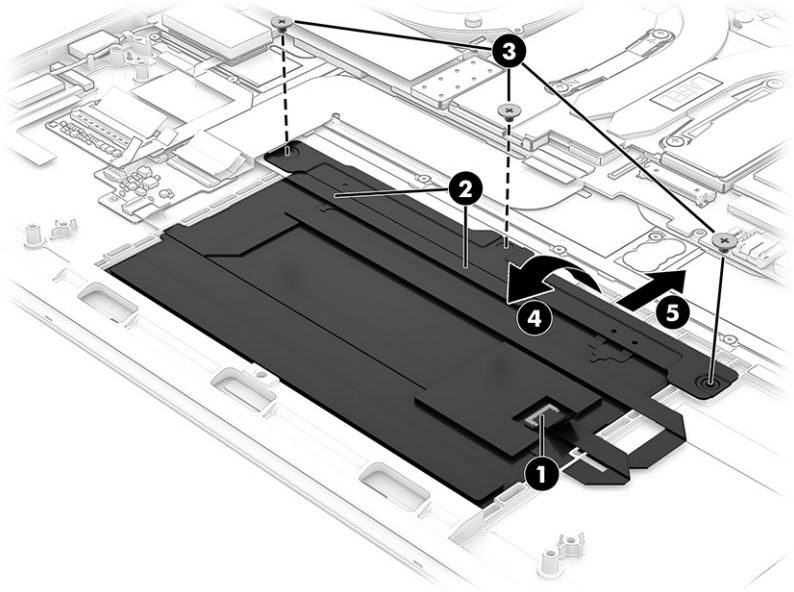
Remove the touchpad:

1. Disconnect the touchpad cable from the touchpad ZIF connector **(1)**.
2. Gently pull up to release the keyboard cable from the long strip of adhesive that secures it to the top of the touchpad **(2)**.

3. Remove the three Phillips M2.0 × 2.0 screws **(3)** that secure the touchpad to the computer.

NOTE: To avoid damaging the display, lift the computer off the display before removing the touchpad. Because you have to rotate the top of the touchpad up at an angle during removal, the bottom can pivot down and damage the display if the touchpad is sitting directly on top of the display.

4. Rotate the touchpad upward to a 20° to 30° angle **(4)**, and then pull the touchpad into the computer at an angle to remove it **(5)**.



To install the touchpad, reverse this procedure.

IMPORTANT: To avoid damaging the display during touchpad installation, before inserting the touchpad into the computer, lift the computer up off the display.

Heat sink

To remove the heat sink, use this procedure and illustration.

Table 5-11 Heat sink description and part number

Description	Spare part number
Heat sink	N22936-001

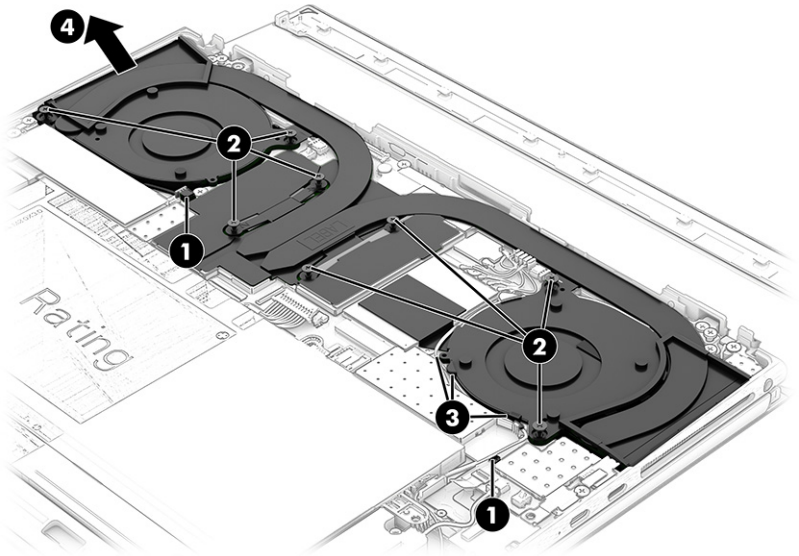
Before removing the heat sink, follow these steps:

1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the bottom cover (see Bottom cover on page 33).
3. Disconnect the battery cable from the system board (see Battery on page 42).

Remove the heat sink:

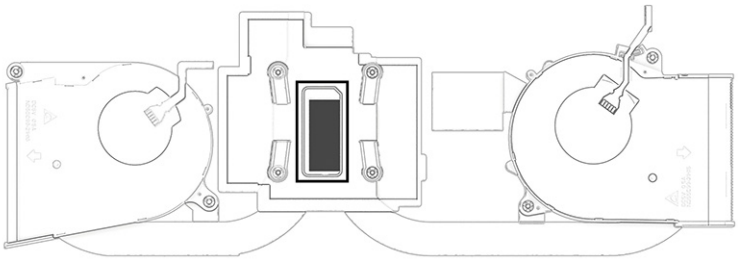
1. Disconnect the two fan cables from the system board **(1)**.

2. Loosen the eight captive Phillips screws **(2)** that secure the heat sink and fans to the computer.
3. Remove the antenna cable from the tape and clips around the right fan **(3)**.
4. Lift the left side of the heat sink up to approximately 45°, and then pull it to the left and out of the computer. **(4)**.



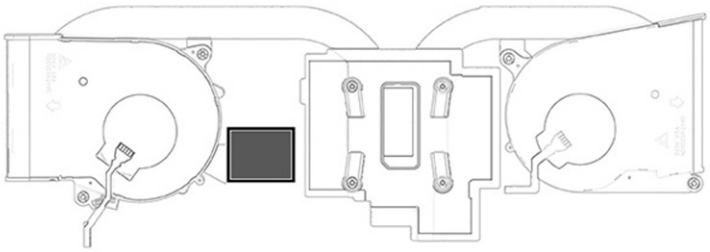
5. Thoroughly clean the thermal material from the surfaces of the heat sink and the system board components each time the heat sink is removed. The following illustration shows the replacement thermal material location.

NOTE: If a thermal pad is not present in replacement heat sink kit, reuse the old thermal pad during the replacement.



To install the heat sink, reverse this procedure.

The replacement heat sink comes with a thermal pad installed on the bottom of the section that extends over the WWAN module.



Top speakers

To remove the top speakers, use this procedure and illustration.

Table 5-12 Top speakers description and part number

Description	Spare part number
Speakers	N22946-001

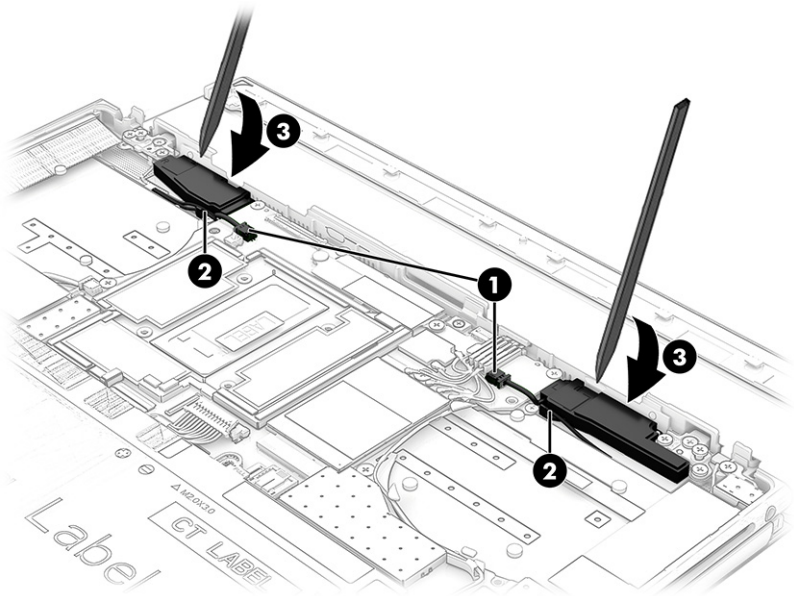
Before removing the top speakers, follow these steps:

1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the top cover (see Bottom cover on page 33).
3. Disconnect the battery cable from the system board (see Battery on page 42).
4. Remove the heat sink (see Heat sink on page 47).

Remove the top speakers:

1. Disconnect the both speaker cables from the system board **(1)**.
2. Remove the antenna cables from the clips in each speaker **(2)**.

3. Use a tool to release each speaker from the computer **(3)**.



To install the top speakers, reverse this procedure.

System board and audio jack

To remove the system board and audio jack, use these procedures and illustrations.

Table 5-13 System board and audio jack descriptions and part numbers

Description	Spare part number
System board (includes integrated processor)	
All system boards use the following part numbers:	
xxxxxx-001: Non-Windows operating systems	
xxxxxx-601: Windows operating system	
Models with integrated vPro WLAN:	
Intel Core i7-1265U processor and 32 GB of system memory	N20728-xxx
Intel Core i7-1265U processor and 16 GB of system memory	N20727-xxx
Intel Core i7-1265U processor and 16 GB of system memory (OSR)	N20726-xxx
Intel Core i5-1245U processor and 16 GB of system memory	N20723-xxx
Intel Core i5-1245U processor and 16 GB of system memory (OSR)	N20722-xxx
Models with integrated non-vPro WLAN:	
Intel Core i7-1255U processor and 32 GB of system memory	N20725-xxx
Intel Core i7-1255U processor and 16 GB of system memory	N20724-xxx
Intel Core i5-1235U processor and 16 GB of system memory	N20720-xxx
Intel Core i5-1235U processor and 8 GB of system memory	N20721-xxx
USB bracket (available in the Bracket Kit)	N22931-001

Table 5-13 System board and audio jack descriptions and part numbers (continued)

Description	Spare part number
Audio jack	N31280-001

Before removing the system board and audio jack, follow these steps:

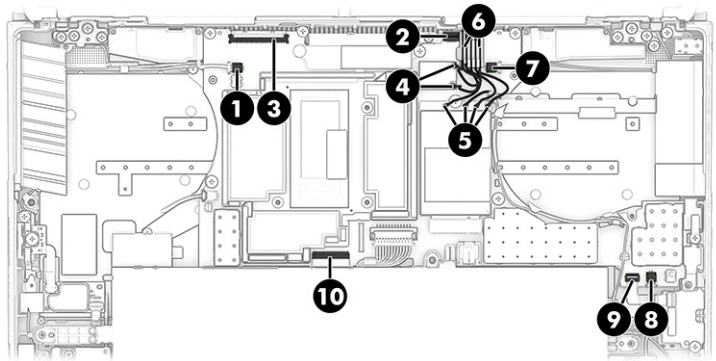
1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the bottom cover (see Bottom cover on page 33).
3. Disconnect the battery cable from the system board (see Battery on page 42).
4. Remove the heat sink (see Heat sink on page 47).

Remove the system board and audio jack:

1. Disconnect the following cables from the system board:
   - Left top speaker cable **(1)**
   - Camera cable **(2)**
   - Display cable **(3)**

   NOTE: Remove one screw and bracket to access the display cable.

   - Antenna cables from onboard WLAN module **(4)**
   - WWAN antennas from the WWAN module **(5)**
   - WWAN antennas from the channels on the system board **(6)**
   - Right top speaker cable **(7)**
   - Audio board cable **(8)**
   - SIM board cable **(9)**
   - Transfer board cable **(10)**



2. Remove the four Phillips M2.0 × 4.0 screws **(1)**.

NOTE: To avoid damage to the computer, be sure to note screw sizes and locations so you can correctly install the correct screws when installing the board.

3. Remove the USB bracket **(2)**.

4. Remove the two Phillips M2.0 × 2.0 screws **(3)**.

NOTE: The screw might be covered by Mylar.

NOTE: To avoid damage to the computer, be sure to note screw sizes and locations so you can correctly install the correct screws when installing the board.

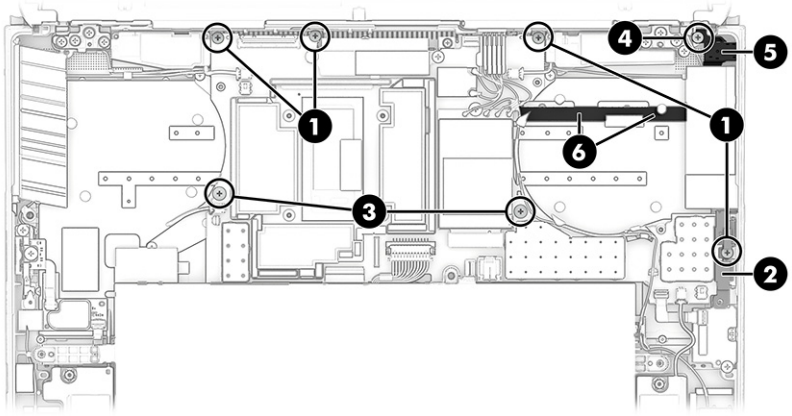
5. Remove the Phillips M2.0 × 4.0 screw **(4)** from the audio jack.

NOTE: To avoid damage to the computer, be sure to note screw sizes and locations so you can correctly install the correct screws when installing the board.

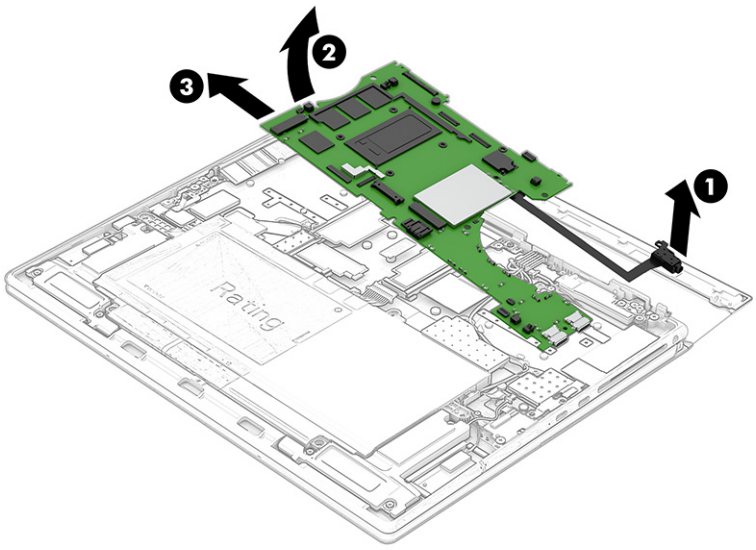
6. Release the audio jack **(5)**.

7. Peel the audio board cable off the computer **(6)**.

NOTE: The audio jack cable connects to the bottom of the system board. It is recommended to leave the cable attached and remove the jack with the system board.



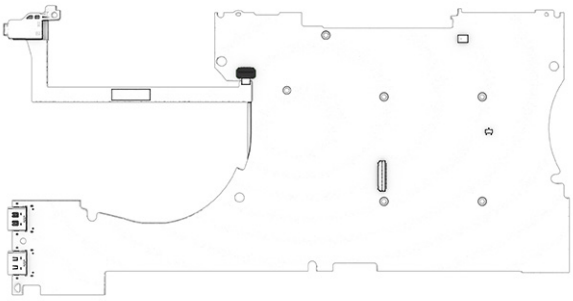
8. With the audio jack and cable connected to the system board **(1)**, lift the left side of the system board up **(2)**, and then pull the system board with the audio back attached toward the left and out of the computer **(3)**.



Reverse this procedure to install the system board.

When replacing the system board, remove the audio jack and cable assembly from the old board and install the assembly on the new board.

The following illustration shows the location of the audio jack cable connector on the bottom of the system board.



Antenna barrel

To remove and disassemble the antenna barrel, use these procedures and illustrations.

Table 5-14 **Antenna barrel description and part number**

Description	Spare part number
Antenna barrel (WWAN)	N22928-001
Antenna barrel (WLAN)	N22927-001
Bumpers (included in Rubber Kit)	N22933-001

Before removing the antenna barrel, follow these steps:

1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the bottom cover (see Bottom cover on page 33).
3. Disconnect the battery cable from the system board (see Battery on page 42).
4. Remove the system board (see System board and audio jack on page 50).

Remove the antenna barrel:

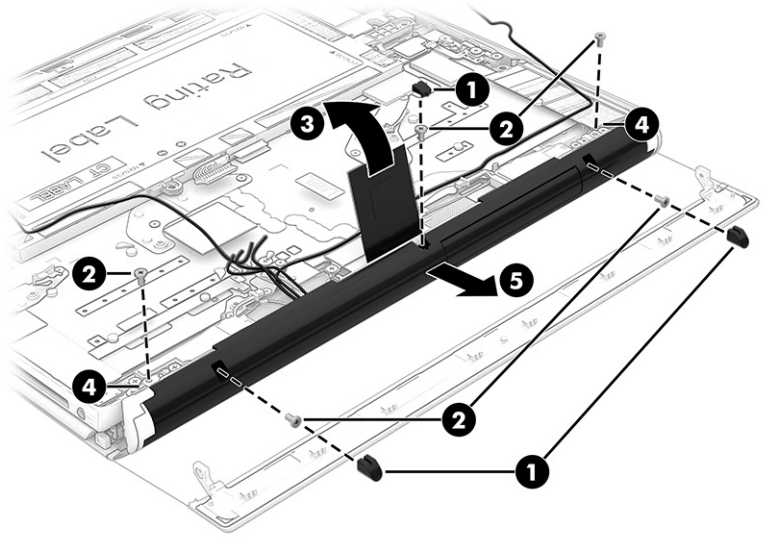
1. Remove the three rubber plugs from the barrel **(1)**.
2. Remove the five Phillips M2.0 × 4.0 screws **(2)** that secure the antenna barrel to the computer.

NOTE: Three of the screws are removed vertically. Two of the screws are removed horizontally.

3. Peel up the orange cable that covers the barrel **(3)**.
4. Remove the antenna barrel cables, from the left and right ends of the barrel, from the clips along the top of the computer **(4)**.

NOTE: If orange Mylar covers each hinge, gently pull up to release the Mylar.

5. Remove the antenna barrel from the computer **(5)**.



Reverse this procedure to install the antenna barrel.

Wireless antennas and cables

To remove the wireless antennas and cables, use this procedure and illustration.

Table 5-15 Wireless antennas and cables description and part number

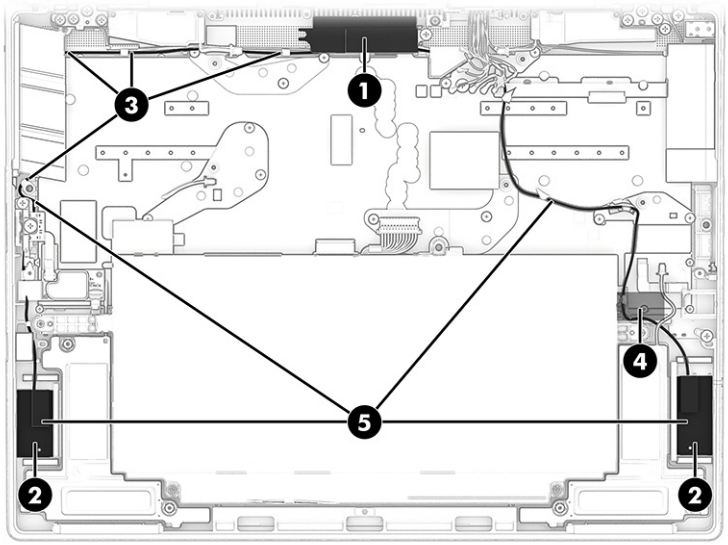
Description	Spare part number
Wireless antennas and cables (aux2)	N22921-001
Wireless antennas and cables (aux3)	N22922-001

Before removing the wireless antennas and cables, follow these steps:

1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the bottom cover (see Bottom cover on page 33).
3. Disconnect the battery cable from the system board (see Battery on page 42).
4. Remove the wireless module (see Wireless charging module on page 41).
5. Remove the heat sink (see Heat sink on page 47).
6. Remove the system board (see System board and audio jack on page 50).

Remove the wireless antennas and cables:

1. Peel up the orange cable that the left antenna cable runs underneath **(1)**.
2. Peel the left and right antennas off the chassis **(2)**.
3. Remove the left antenna cable from the clips along the side and top of the computer **(3)**.
4. Remove the tape next to the SIM board that secures the right cable to the chassis **(4)**.
5. Remove the antennas and cables from the computer **(5)**.



To install the wireless antennas and cables, reverse this procedure.

Top cover with keyboard

The top cover with keyboard remains after removing all other spare parts from the computer. In this section, the first table provides the main spare part number for the top cover with keyboards. The second table provides the country codes.

Table 5-16 Top cover with keyboard descriptions and part numbers

Description	Spare part number
Top cover with keyboard, privacy models (includes thermal grease and replacement antennas)	N22552-xxx
Top cover with keyboard, nonprivacy models with an antiglare display (includes thermal grease and replacement antennas)	N24787-xxx
Top cover with keyboard, nonprivacy models with a BrightView display (includes thermal grease and replacement antennas)	N22553-xxx

Table 5-17 Spare part country codes

For use in country or region	Spare part number	For use in country or region	Spare part number	For use in country or region	Spare part number
Belgium	-A41	Hungary	-211	Saudi Arabia	-171
Brazil	-201	Iceland	-DD1	Slovenia	-BA1
Bulgaria	-261	India	-D61	South Korea	-AD1
Chile	-161	Israel	-BB1	Spain	-071
Czech Republic/Slovakia	-FL1	Italy	-061	Switzerland	-BG1
Denmark	-081	Japan	-291	Taiwan	-AB1
Denmark, Finland, and Norway	-DH1	The Netherlands	-B31	Thailand	-281
French Canada	-DB1	Northern Africa	-FP1	Turkey	-141
Finland/Sweden	-B71	Norway	-091	Turkey-F	-541
France	-051	Portugal	-131	Ukraine	-BD1
Germany	-041	Romania	-271	United Kingdom	-031
Greece	-151	Russia	-251	United States	-001

Display panel and subcomponents

To remove the display panel and components, use these procedures and illustrations.

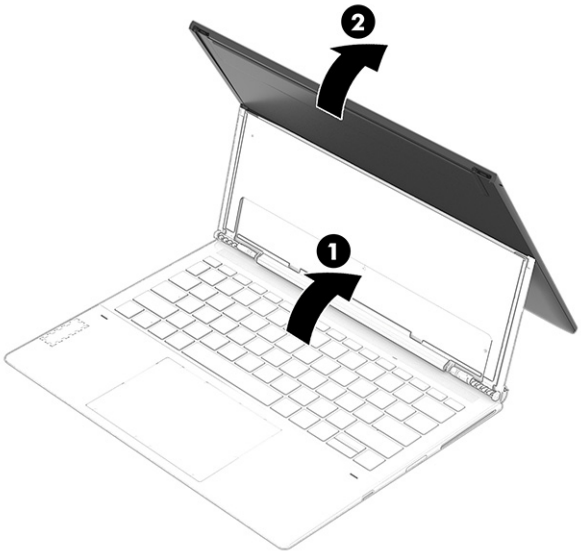
Before removing the display panel, follow these steps:

1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the bottom cover (see Bottom cover on page 33).
3. Disconnect the battery cable from the system board (see Battery on page 42).

Remove the display panel and components:

1. To remove the **display panel**:

a. Open the computer **(1)**, and then flip the display so that it faces away from the keyboard **(2)**.

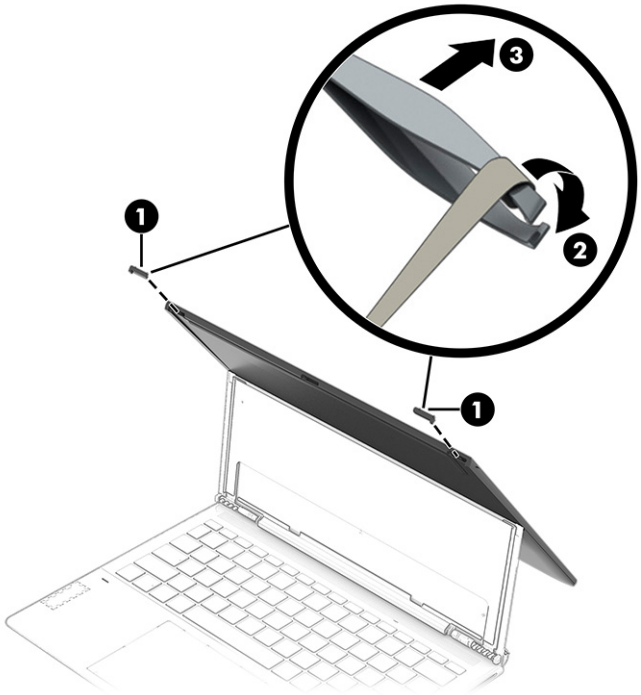


b. Remove the rubber plugs from the left and right sides of the bottom edge of the display **(1)**.

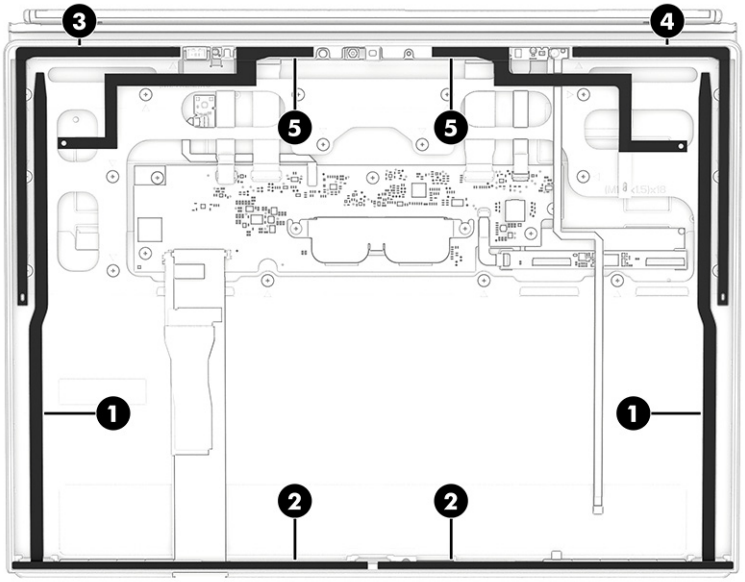
NOTE: There are eight pieces of tape you must remove to remove the display panel.

c. Under the plugs, use tweezers to grasp and pull the end of the display panel tape out from behind the left and right sides of the panel. While turning the tweezers, wrap the tape around the tweezers **(2)** as you continue to pull the tape out from behind the display panel **(3)**. You must pull the tape multiple times before it is completely removed.

IMPORTANT: Slowly and carefully wrap the tape around the tweezers. Pulling on the tape can break it before it is completely removed. Use this process for removing all display tape.



The following illustration shows tape locations. In this step, you are removing the tape shown by callout 1.

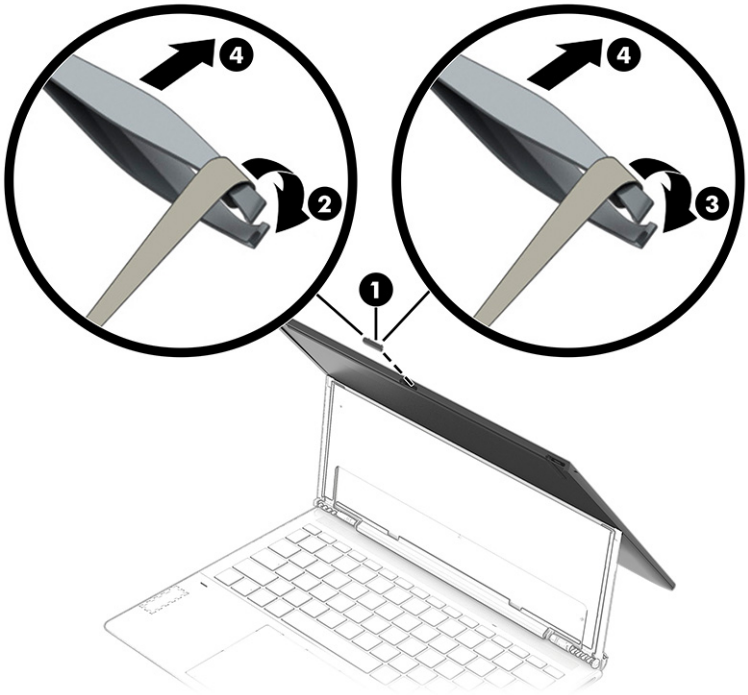


d. Remove the rubber plug from the middle of the bottom edge of the display **(1)**.

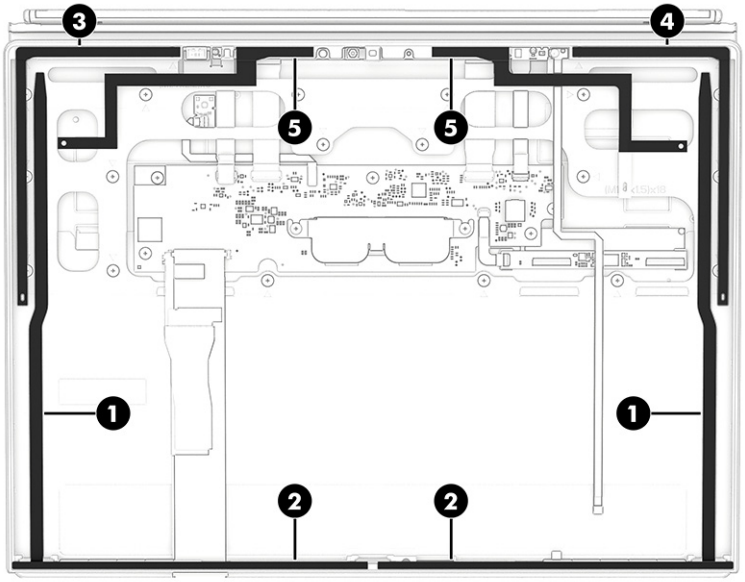
e. Note that tape routes to the left and right from the middle plug. Use tweezers to grasp the tape, and then while turning the tweezers, wrap the tape around the tweezers from the left side of the panel **(2)** as you continue to pull the tape out from behind the display panel **(4)**. Then wrap the tape

around the tweezers from the right side of the panel **(3)** as you continue to pull the tape out from behind the display panel **(4)**.

NOTE: After you grasp the tape with the tweezers, do not pull the tape. Instead, roll the tape around the tweezers while keeping them in the same location. The tape might tear if you pull it.

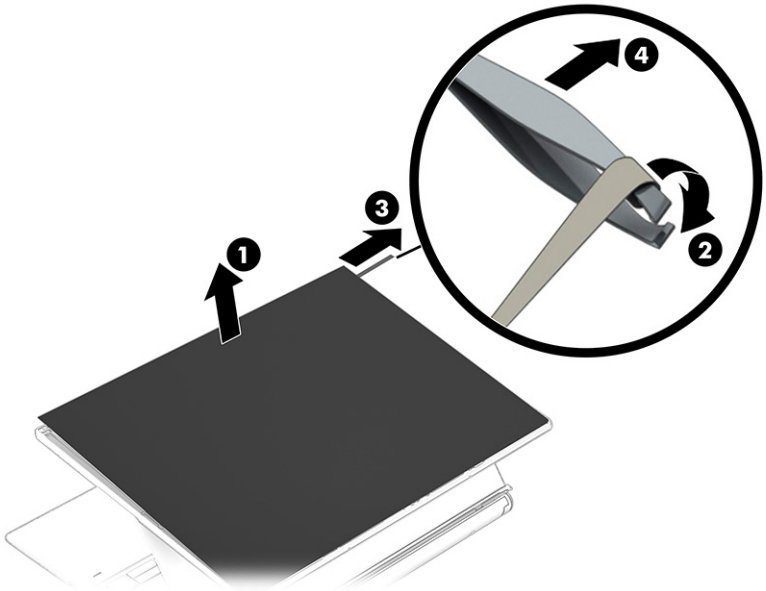


The following illustration shows tape locations. In this step, you are removing the tape shown by callout 2.

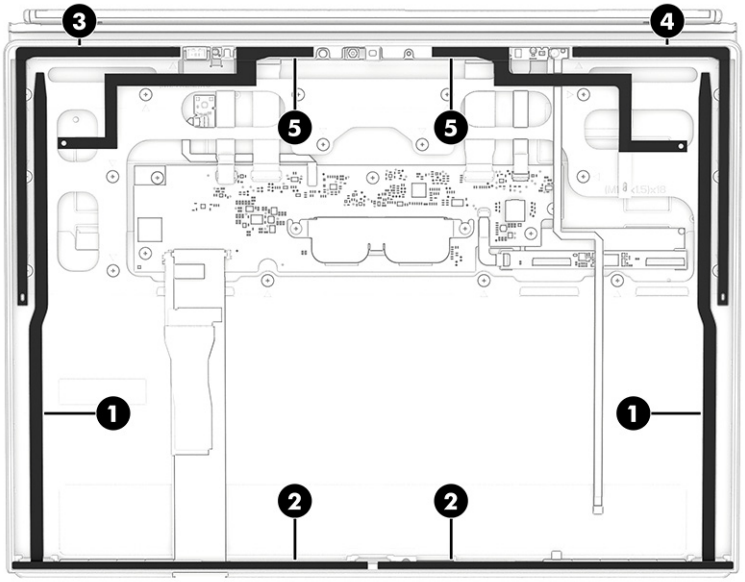


f. Pull the panel away from the display slightly, enough to access more tape underneath **(1)**.

g. Use tweezers **(2)** to grasp and pull the end of the tape out from behind the top right of the panel so that it is perpendicular to the panel **(3)**. While turning the tweezers, wrap the tape around the tweezers as you continue to pull the tape out from behind the display panel **(4)**. You must pull the tape multiple times before it is completely removed.



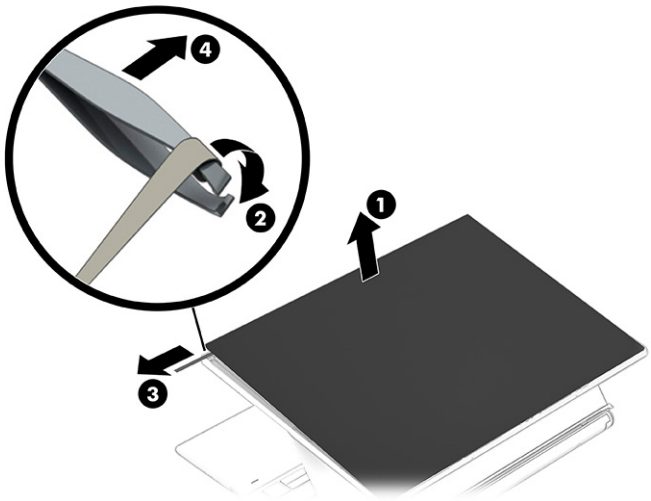
The following illustration shows tape locations. In this step, you are removing the tape shown by callout 3.



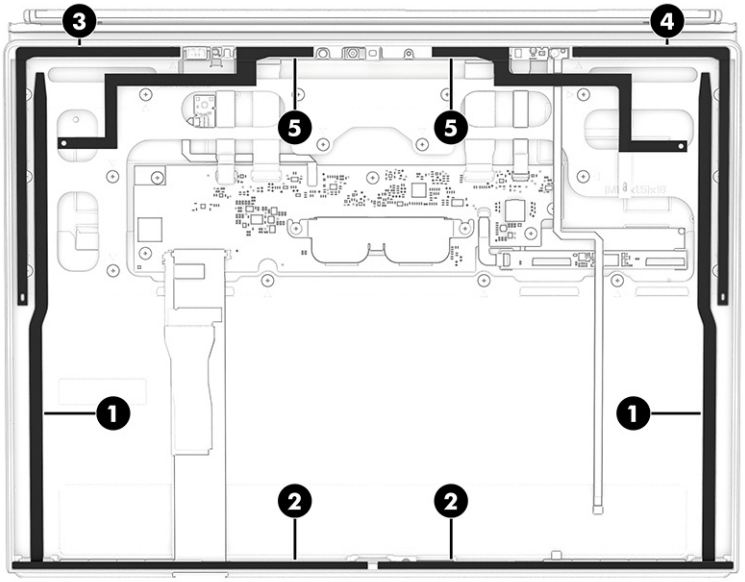
h. Pull the panel away from the display slightly, enough to access more tape underneath **(1)**.

i. Use tweezers **(2)** to grasp and pull the end of the tape out from behind the top left of the panel so that it is perpendicular to the panel **(3)**. While turning the tweezers, wrap the tape around the

tweezers as you continue to pull the tape out from behind the display panel **(4)**. You must pull the tape multiple times before it is completely removed.



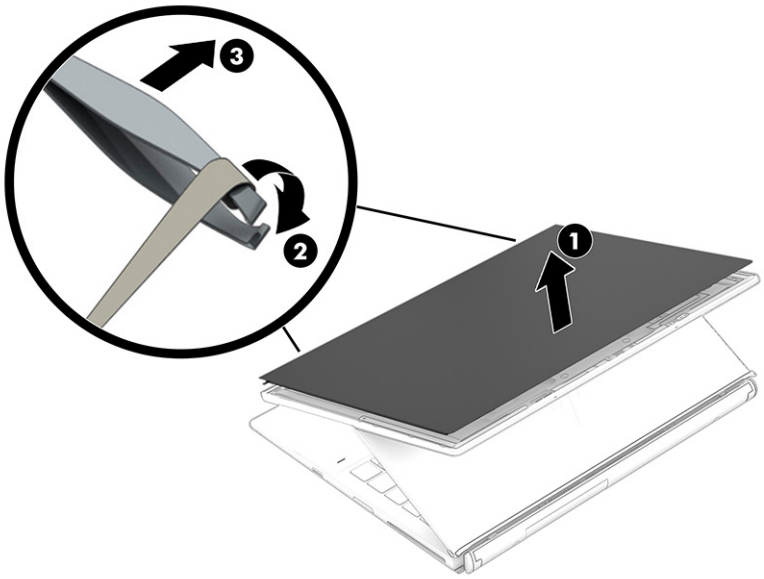
The following illustration shows tape locations. In this step, you are removing the tape shown by callout 4.



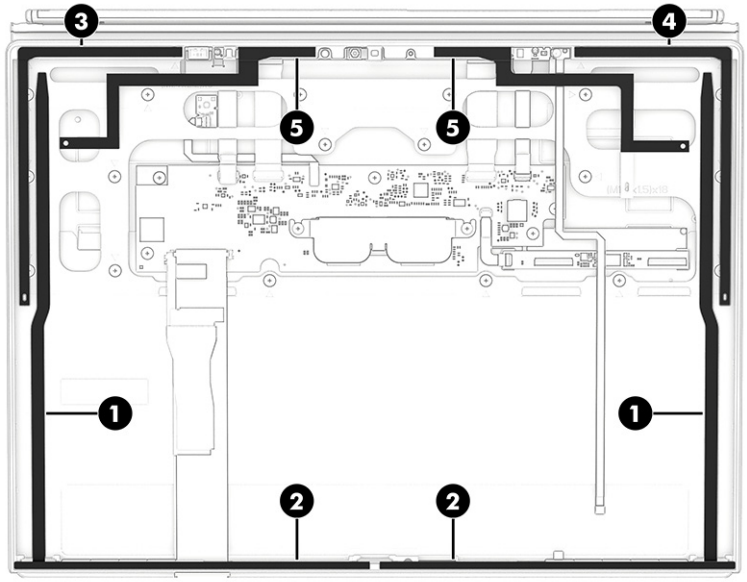
j. Pull the panel away from the display slightly, enough to access more tape underneath **(1)**.

k. Use tweezers to grasp and pull the end of two pieces of tape out from behind the panel so that it is perpendicular to the panel. While turning the tweezers, wrap the tape around the tweezers **(2)** as you continue to pull the tape out from behind the display panel **(3)**. You must pull the tape multiple times before it is completely removed.

NOTE: If the tape tears during removal, you can use a pry tool along the edge of the panel to release it from the computer.



The following illustration shows tape locations. In this step, you are removing the tape shown by callout 5.



l. Separate the panel from the display and place it next to the display rear cover.



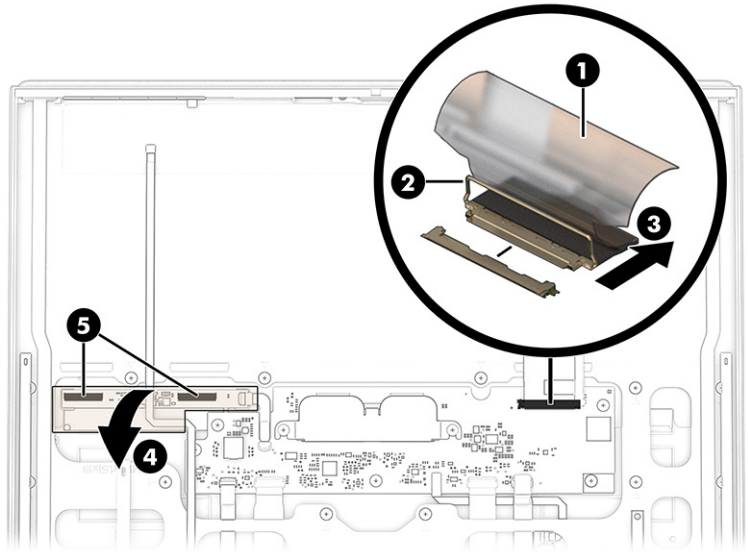
m. Disconnect the panel cables by removing the tape from the left connector **(1)**.

n. Rotate the locking lever **(2)**.

o. Pull the cable out of the connector **(3)**.

p. Remove the foil tape from the right two connectors **(4)**.

q. Disconnect the cables from the two ZIF connectors on the board **(5)**.



Display panels are available as the following spare part numbers:

N23302-001: WUXGA, antiglare

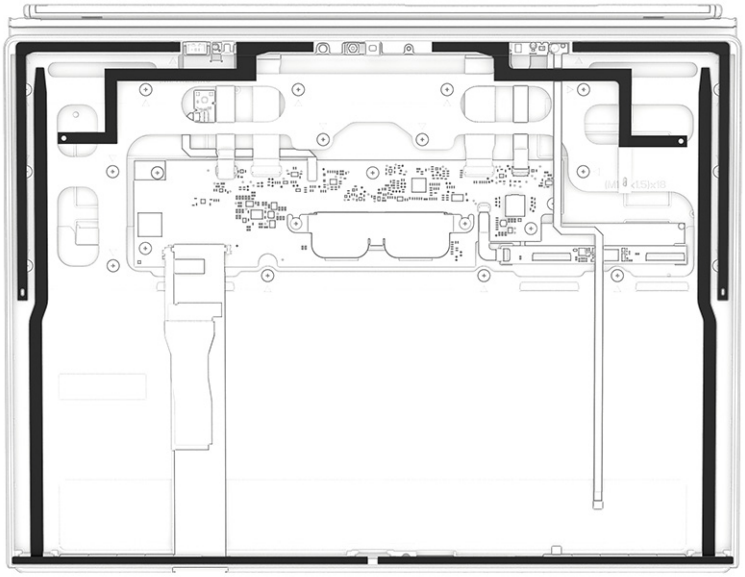
N23304-001: WUXGA, BrightView

N23303-001: WUXGA, privacy

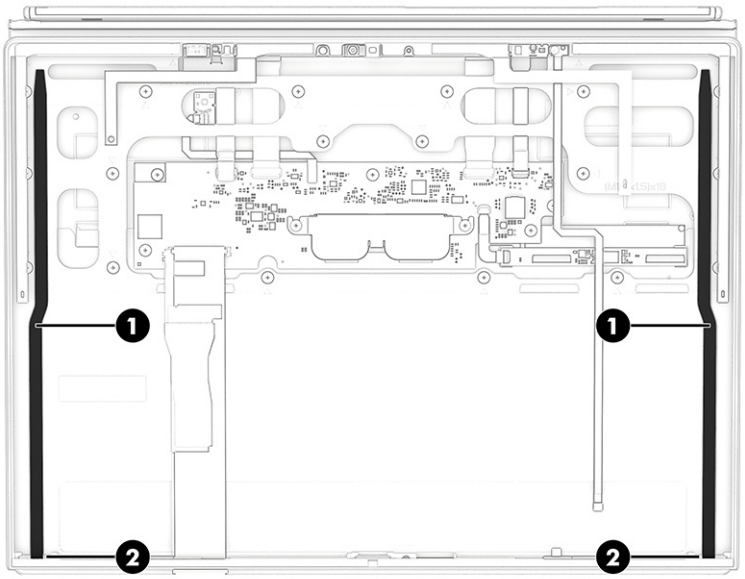
N23305-001: OLED

r. When installing a display panel, eight pieces of tape secure the panel. The following illustration shows their locations.

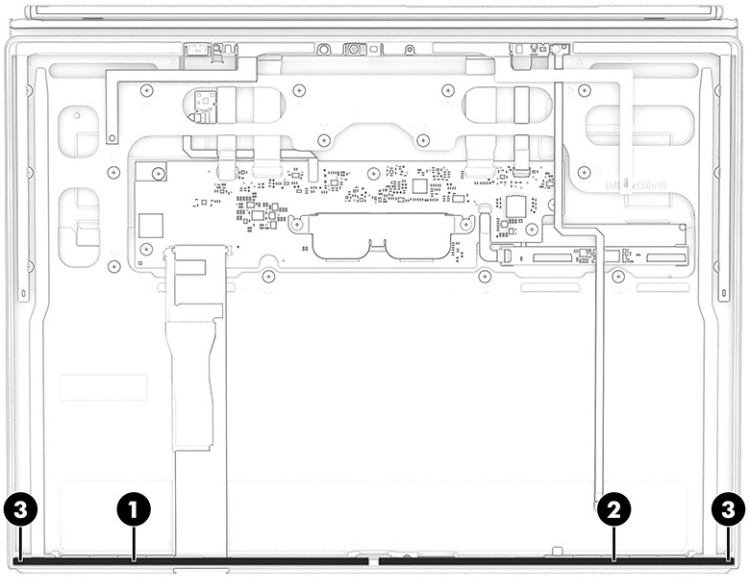
NOTE: The tape is double sided, with plastic covering both sides. Be sure to pull the plastic backing off the bottom when installing the tape onto the display rear cover and off the top when installing the display panel.



s. To install a display panel, install tape at the left and right **(1)** locations as shown in the following illustration. Make sure the tape protrudes enough out of the rubber plug holes on both sides **(2)**.



t. Install tape at the left bottom bezel location **(1)** and the right bottom bezel location **(2)** as shown in the following illustration. When installing the tape, first place the tape in the bottom corners **(3)** of the display and work your way toward the middle.



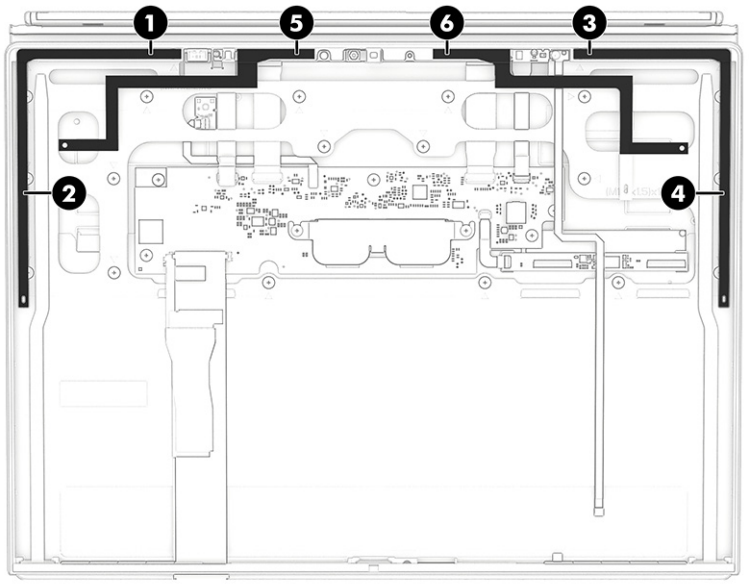
u. Install tape at the following locations:

(1) Top left bezel, starting at right side and working left and down **(2)**.

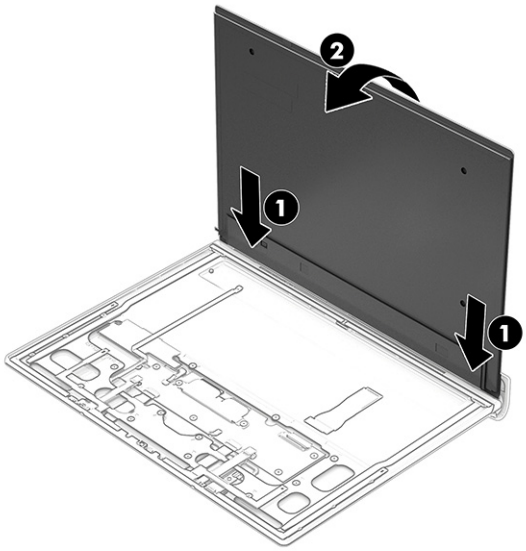
(3) Top right bezel, starting at left side and working right and down **(4)**.

(5) Top left middle, starting at right side and working left.

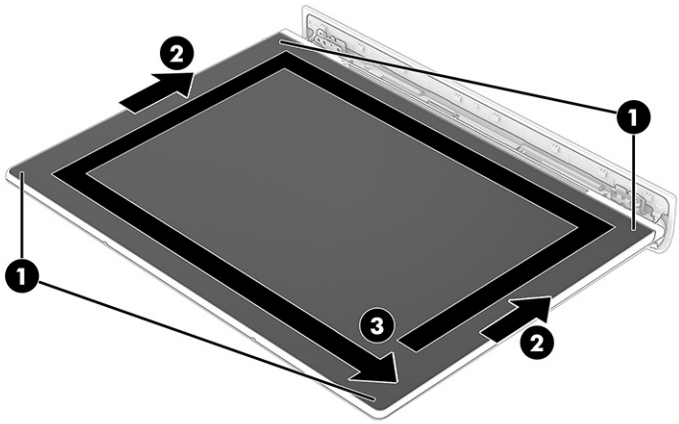
(6) Top right middle, starting at left side and working right.



v. To install a display panel, insert the top two corners of the display panel into the display rear cover **(1)**, and then rotate the panel down into place in the cover **(2)**.



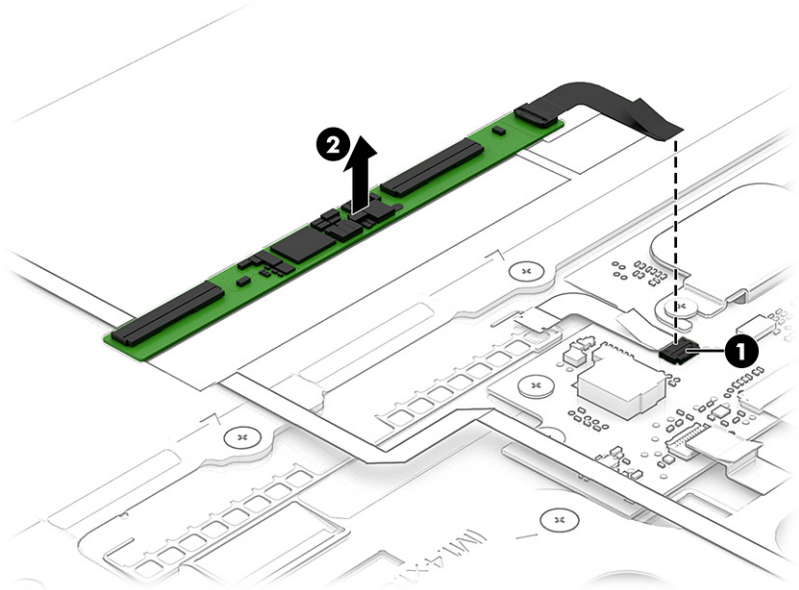
w. To secure the panel to the tape, press down along the right and left sides and across the bottom of the panel **(1)**. Then press about an inch inside along the left and right sides of the panel **(2)**. Finally, press around all four edges of the panel **(3)**.



2. To remove the **touch control board** from the display rear cover:

a. Disconnect the cable from the ZIF connector **(1)**.

b. Peel the board off the inside of the display rear cover **(2)**. The board is secured with adhesive.



The touch control board is available as spare part number N22944-001.

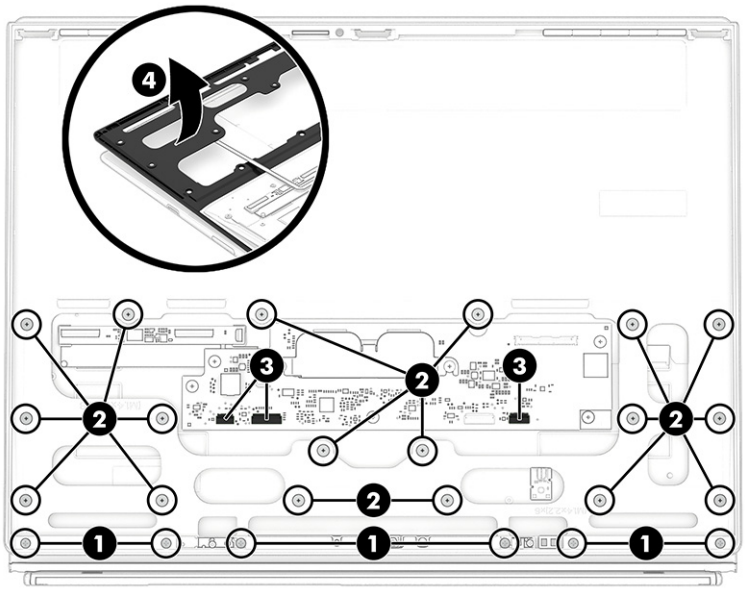
3. To remove the **mid frame** from the display rear cover:

   a. Remove the 6 Phillips 1.4 × 2.2 screws from the bottom edge of the frame **(1)**.

   b. Remove the 18 Phillips 1.4 × 1.5 screws from the remainder of the frame **(2)**.

   c. Disconnect the three cables from the ZIF connectors on the board **(3)**.

   d. Lift the frame up to access the components underneath **(4)**.



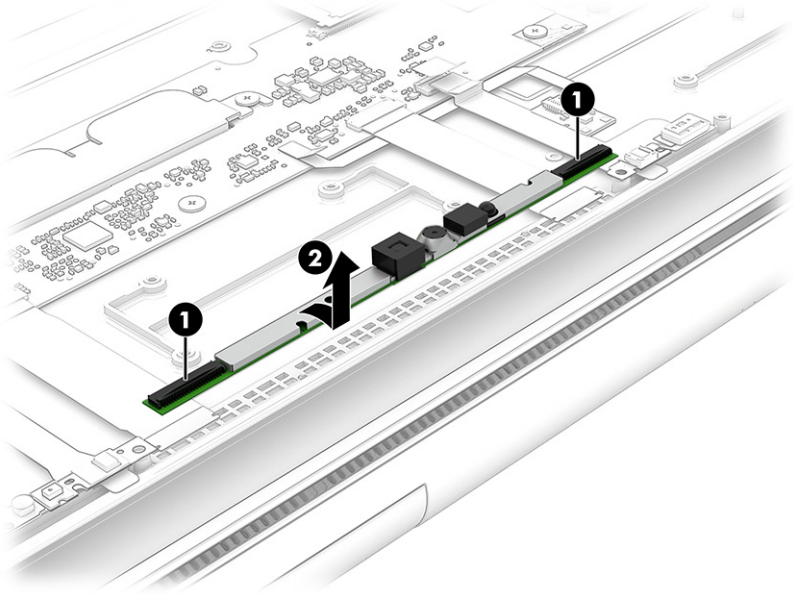
The LCD mid frame is available as spare part number N22923-001 in models with an antiglare display and N22924-001 in models with a BrightView display.

4. To remove the **camera module** from the display rear cover:

   a. Disconnect the two cables from the ZIF connectors on the camera module **(1)**.

   b. Roll the module up to release it **(2)**. The module is secured with adhesive.

   The camera module is available as spare part number N23306-001.



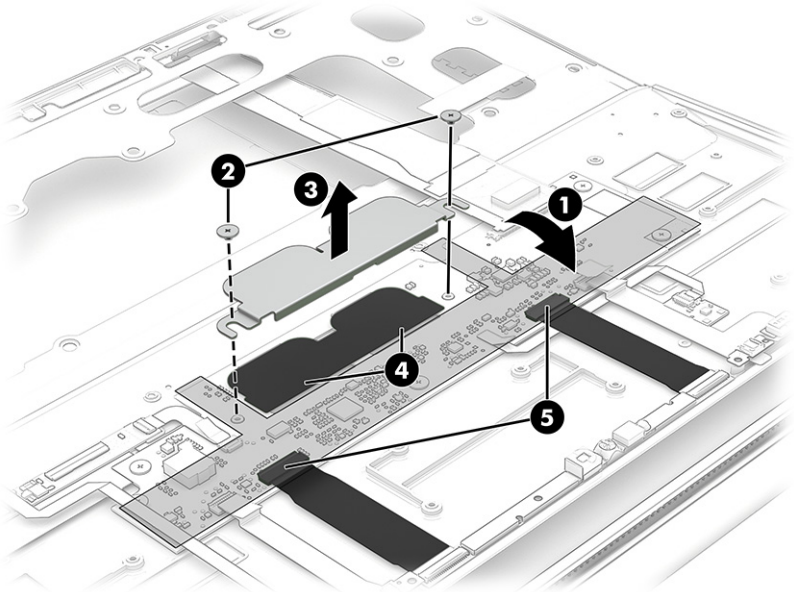
5. To remove the **hub board** from the display rear cover:

   a. Peel the tape off the connectors **(1)**.

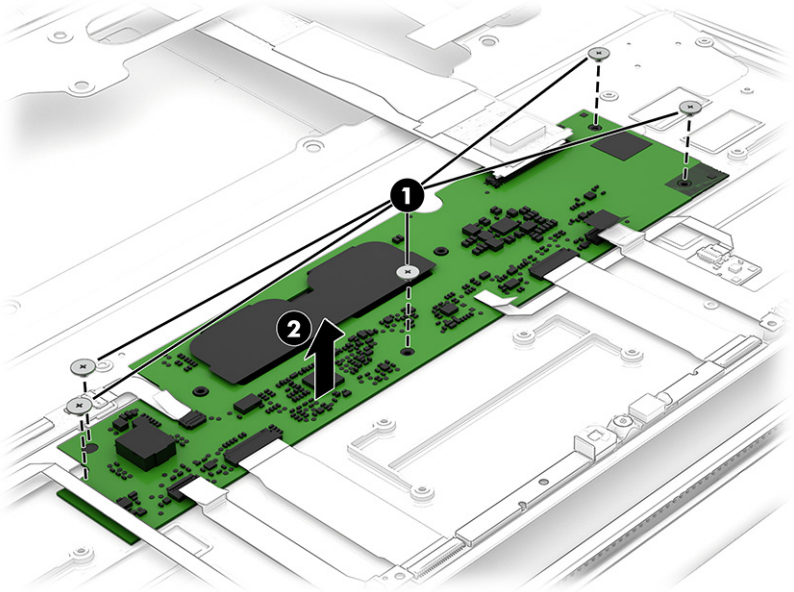
   b. Remove the two Phillips M 2.0 × 3.0 screws **(2)** from the bracket, and then remove the bracket **(3)**.

   c. Disconnect the cables that were under the bracket **(4)**.

   d. Disconnect the two camera module cables from the ZIF connectors on the board **(5)**.

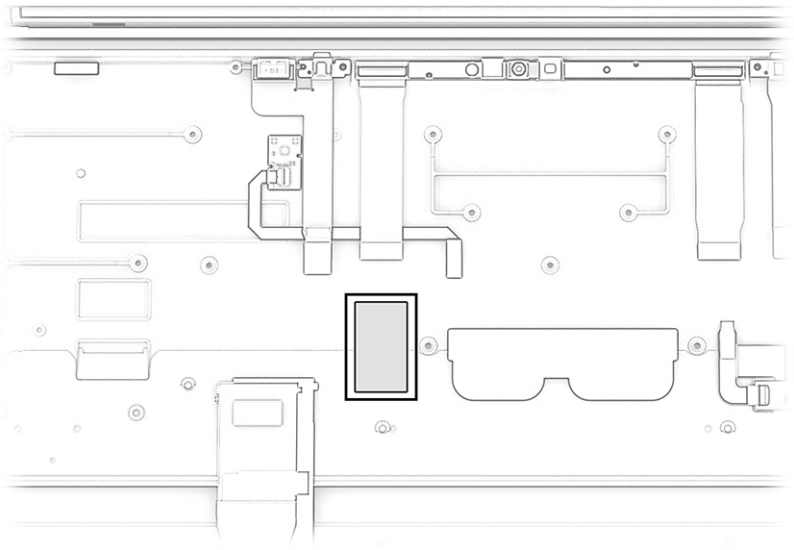


e. Remove the four Phillips M 2.0 × 2.0 screws **(1)**, and then remove the board **(2)**.



f. When installing a hub board, be sure a thermal pad is installed under the board as shown in the following illustration.

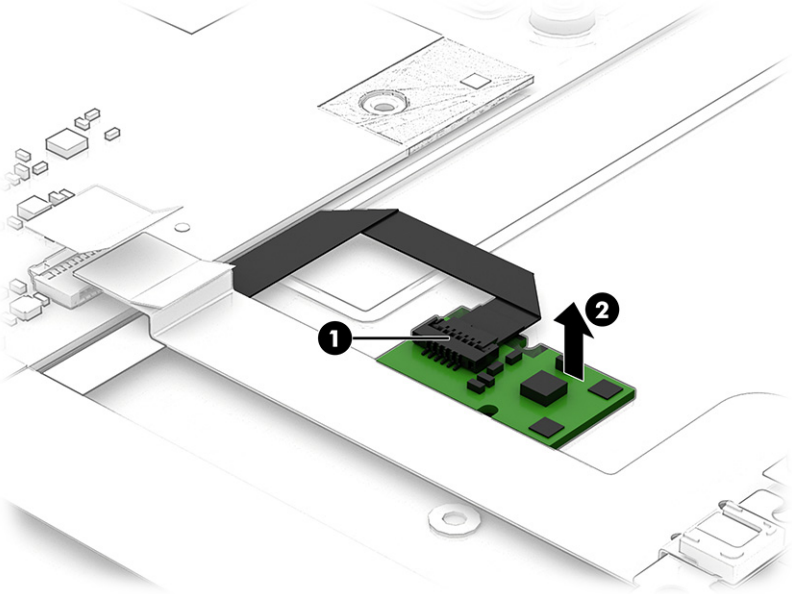
The hub board is available as spare part number N22937-001 for models with FHD panels and N22938-001 for models with OLED panels.



6. To remove the **EC board** from the display rear cover:

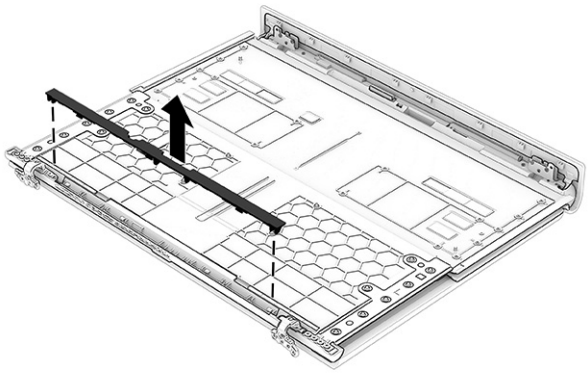
a. Disconnect the cable from the ZIF connector on the board **(1)**.

b. Release the board from the display **(2)**.

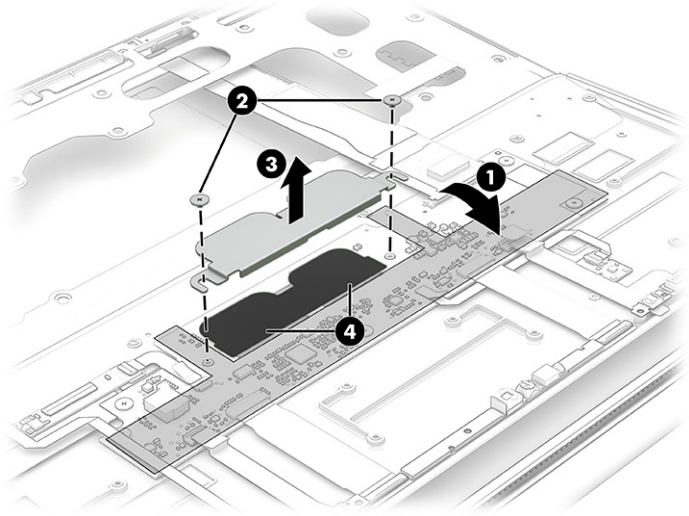


7. To remove the **display cables** from the display rear cover:

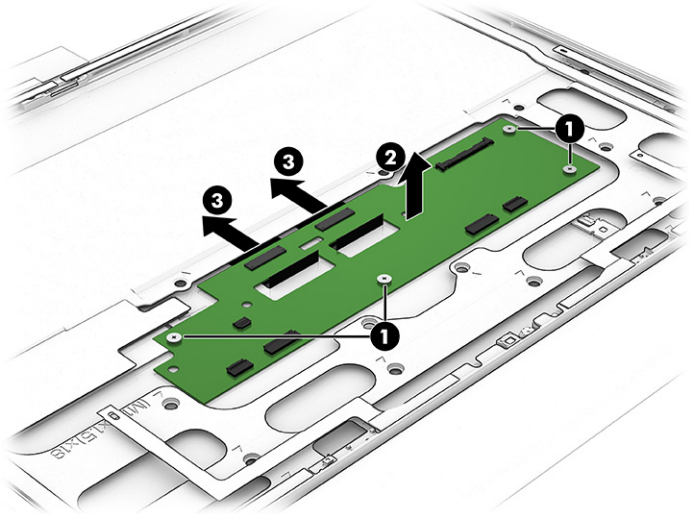
   a. Remove the hinge cover.



b. Peel the tape off the connectors **(1)**. Remove the two Phillips M 2.0 × 3.0 screws **(2)** from the bracket, and then remove the bracket **(3)**. Disconnect the cables that were under the bracket **(4)**.

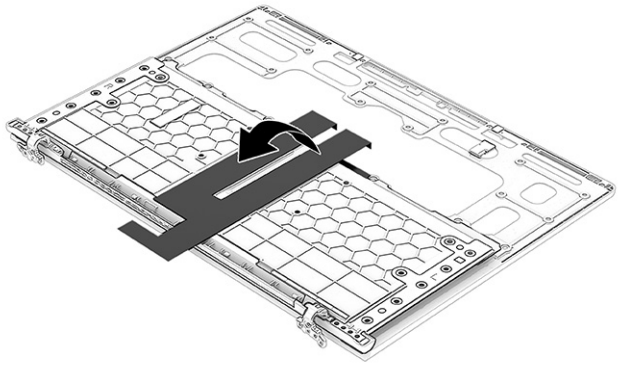


c. Loosen the four Philips M2.0 × 2.0 screws **(1)** from the system board. Lift the board up slightly **(2)**, and then pull the cables out through the holes in the board **(3)**.



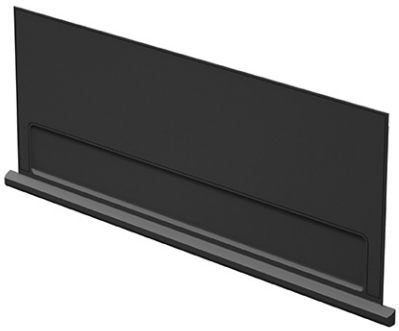
d. Peel the display cables off the computer.

Display cables are available in the Display Cable Kit as spare part number N22930-001.



8. After all parts are removed, the metal back cover for the hinge arm remains.

   The metal back cover for the hinge arm is available in the Bracket Kit as spare part number N22925-001.



Reverse this procedure to install the display panel.

Display assembly

To remove the display assembly, use these procedures and illustrations. You must remove the display assembly and the display panel to access the display cables and the back cover with leather cover.

NOTE: Only remove the display assembly if you need to replace the hinges, display cables, or back cover with leather cover.

To access the display cables and back cover with leather cover, you must also remove the display panel (see Display panel and subcomponents on page 56).

NOTE: Spare parts for displays are available only at the subcomponent level.

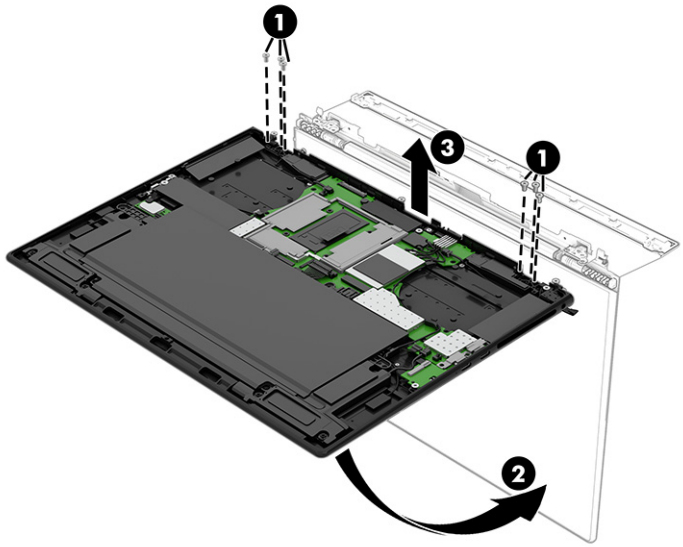
Before removing the display panel, follow these steps:

1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the bottom cover (see Bottom cover on page 33).

3. Disconnect the battery cable from the system board (see Battery on page 42).
4. Remove the system board (see System board and audio jack on page 50).
5. Remove the antenna barrel (see Antenna barrel on page 53).

Remove the display assembly:

- To remove the display assembly:
  a. Remove the three Phillips M2.0 × 4.0 screws **(1)** from each hinge.
  b. Open the computer **(2)**, and then separate the computer from the display **(3)**.



Reverse this procedure to install the display assembly.

Display hinges

To remove the display hinges, use these procedures and illustrations. The display back cover with leather (spare part number N22920-001) remains after removing all other spare parts from the display.

Before removing the display hinges, follow these steps:

1. Prepare the computer for disassembly (see Preparation for disassembly on page 33).
2. Remove the bottom cover (see Bottom cover on page 33).
3. Disconnect the battery cable from the system board (see Battery on page 42).
4. Remove the display panel (see Display panel and subcomponents on page 56).
5. Remove the system board (see System board and audio jack on page 50).
6. Remove the antenna barrel (see Antenna barrel on page 53).
7. Remove the display assembly from the computer (see Display assembly on page 72).

Remove the display hinges:

NOTE: To avoid damaging the display, place the panel on soft material during disassembly.

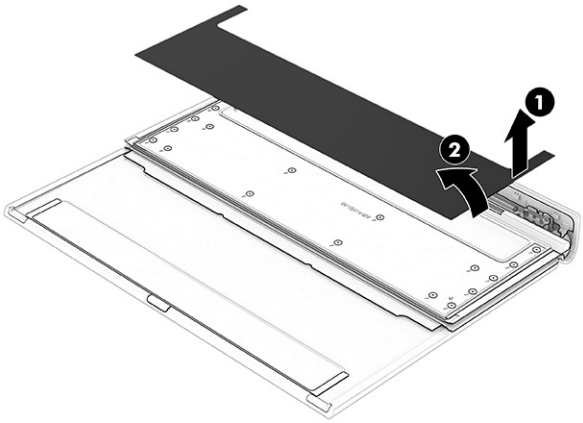
1. To remove the hinges:

   a. Position the removed display panel as shown in the following illustration.

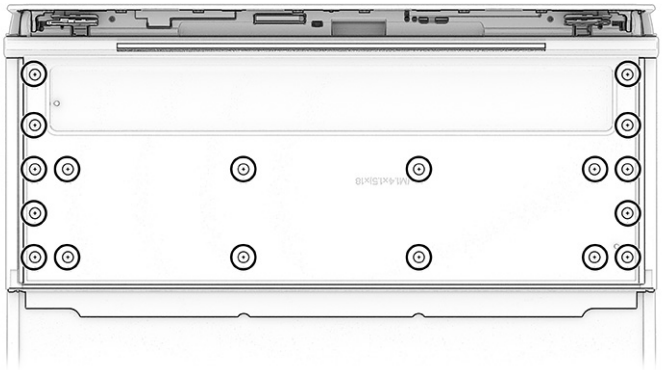


   b. Release the edge of the Mylar **(1)**, and then peel the large piece of Mylar off the display enclosure **(2)**.

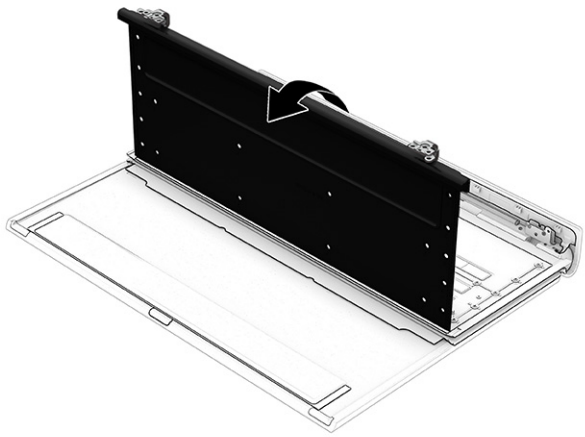
   NOTE: Take care not to tear the Mylar as you will reuse it during reassembly.



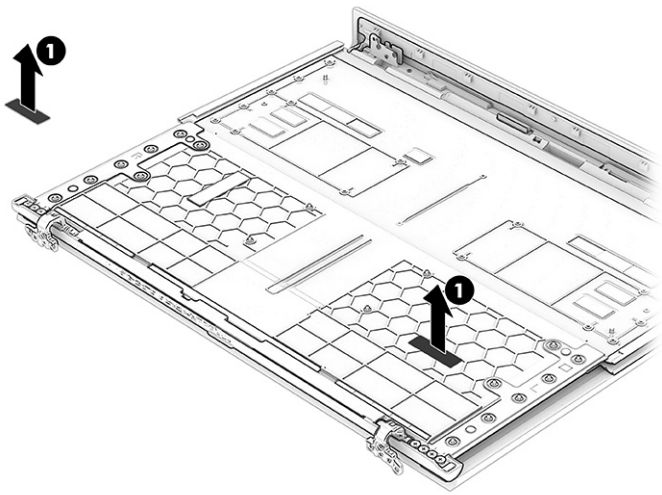
c. Remove the 18 Phillips 1.4 × 1.5 screws that are located under the removed Mylar.



d. Rotate the enclosure over to access the hinges.



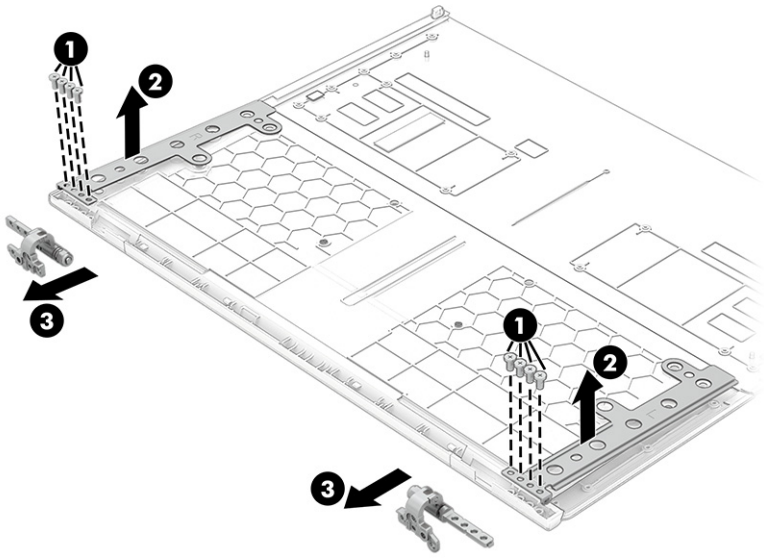
e. Remove the Mylar from the screws on both hinges **(1)**.



f. Remove the four Phillips 2.0 × 3.0 screws **(1)** from each hinge.

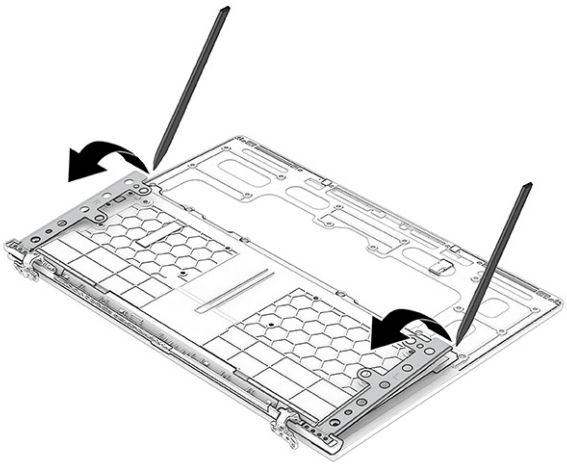
g. Lift the plates **(2)**, and then pull the hinges out from under the plates **(3)**.

The hinges are available as spare part number N22935-001 for FHD displays and N26147-001 for OLED displays.



2. To remove the hinge arm support brackets, use a tool to release the brackets from the computer. The brackets are secured with adhesive.

Hinge arm support brackets are available in the Bracket Kit as spare part number N22931-001.



Reverse this procedure to install the display hinges.

6 Backing up, restoring, and recovering

You can use Windows tools or HP software to back up your information, create a restore point, reset your computer, create recovery media, or restore your computer to its factory state. Performing these standard procedures can return your computer to a working state faster.

IMPORTANT: If you are performing recovery procedures on a tablet, the tablet battery must be at least 70% charged before you start the recovery process.

IMPORTANT: For a tablet with a detachable keyboard, connect the tablet to the keyboard base before beginning any recovery process.

Backing up information and creating recovery media

These methods of creating recovery media and backups are available on select products only.

Using Windows tools for backing up

HP recommends that you back up your information immediately after initial setup. You can do this task either using Windows Backup locally with an external USB flash drive or using online tools.

IMPORTANT: Windows is the only option that allows you to back up your personal information. Schedule regular backups to avoid information loss.

NOTE: If computer storage is 32 GB or less, Microsoft® System Restore is disabled by default.

Using the HP Cloud Recovery Download Tool to create recovery media (select products only)

You can use the HP Cloud Recovery Download Tool to create HP Recovery media on a bootable USB flash drive.

For details:

- Go to http://www.hp.com, search for HP Cloud Recovery, and then select the result that matches the type of computer that you have.

NOTE: If you cannot create recovery media yourself, contact support to obtain recovery discs. Go to http://www.hp.com/support, select your country or region, and then follow the on-screen instructions.

IMPORTANT: HP recommends that you follow the Restoring and recovery methods on page 78 to restore your computer before you obtain and use the HP recovery discs. Using a recent backup can return your machine to a working state sooner than using the HP recovery discs. After the system is restored, reinstalling all the operating system software released since your initial purchase can be a lengthy process.

Restoring and recovering your system

You have several tools available to recover your system both within and outside of Windows if the desktop cannot load.

HP recommends that you attempt to restore your system using the Restoring and recovery methods on page 78.

Creating a system restore

System Restore is available in Windows. The System Restore software can automatically or manually create restore points, or snapshots, of the system files and settings on the computer at a particular point.

When you use System Restore, it returns your computer to its state at the time you made the restore point. Your personal files and documents should not be affected.

Restoring and recovery methods

After you run the first method, test to see whether the issue still exists before you proceed to the next method, which might now be unnecessary.

1. Run a Microsoft System Restore.
2. Run Reset this PC.

   NOTE: The options **Remove everything** and then **Fully clean the drive** can take several hours to complete and leave no information on your computer. It is the safest way to reset your computer before you recycle it.

3. Recover using HP Recovery media. For more information, see Recovering using HP Recovery media on page 78.

For more information about the first two methods, see the Get Help app:

- Select the **Start** button, select **All apps**, select the **Get Help** app, and then enter the task you want to perform.

NOTE: You must be connected to the internet to access the Get Help app.

Recovering using HP Recovery media

You can use HP Recovery media to recover the operating system and drivers that were installed at the factory. On select products, you can create recovery media on a bootable USB flash drive using the HP Cloud Recovery Download Tool.

For details, see Using the HP Cloud Recovery Download Tool to create recovery media (select products only) on page 77.

NOTE: If you cannot create recovery media yourself, contact support to obtain recovery discs. Go to http://www.hp.com/support, select your country or region, and then follow the on-screen instructions.

To recover your system:

- Insert the HP Recovery media, and then restart the computer.

NOTE: HP recommends that you follow the Restoring and recovery methods on page 78 to restore your computer before you obtain and use the HP recovery discs. Using a recent backup can return your machine to a working state sooner than using the HP recovery discs. After the system is restored, reinstalling all the operating system software released since your initial purchase can be a lengthy process.

Changing the computer boot order

If your computer does not restart using the HP Recovery media, you can change the computer boot order, which is the order of devices listed in BIOS for startup information. You can select an optical drive or a USB flash drive, depending on the location of your HP Recovery media.

IMPORTANT: For a tablet with a detachable keyboard, connect the tablet to the keyboard base before beginning these steps.

To change the boot order:

1. Insert the HP Recovery media.
2. Access the system **Startup** menu.
   - For computers or tablets with keyboards attached, turn on or restart the computer or tablet, quickly press esc, and then press f9 for boot options.
   - For tablets without keyboards, turn on or restart the tablet, quickly press and hold the volume up button, and then select **f9**.

     – or –

     Turn on or restart the tablet, quickly press and hold the volume down button, and then select **f9**.
3. Select the optical drive or USB flash drive from which you want to boot, and then follow the on-screen instructions.

Using HP Sure Recover (select products only)

Select computer models are configured with HP Sure Recover, a PC operating system (OS) recovery solution built into the hardware and software. HP Sure Recover can fully restore the HP OS image without installed recovery software.

Using HP Sure Recover, an administrator or user can restore the system and install:

- Latest version of the operating system
- Platform-specific device drivers
- Software applications, in the case of a custom image

To access the latest documentation for HP Sure Recover, go to http://www.hp.com/support. Follow the on-screen instructions to find your product and locate your documentation.

7 Computer Setup (BIOS), TPM, and HP Sure Start

HP provides several tools to help set up and protect your computer.

Using Computer Setup

Computer Setup, or Basic Input/Output System (BIOS), controls communication between all the input and output devices on the system (such as hard drives, display, keyboard, mouse, and printer). Computer Setup includes settings for types of devices installed, the startup sequence of the computer, and amount of system and extended memory.

NOTE: Use extreme care when making changes in Computer Setup. Errors can prevent the computer from operating properly.

To start Computer Setup, turn on or restart the computer, and when the HP logo appears, press f10 to enter Computer Setup.

Navigating and selecting in Computer Setup

You can navigate and select in Computer Setup using one or more methods.

- To select a menu or a menu item, use the tab key and the keyboard arrow keys and then press enter, or use a pointing device to select the item.
- To scroll up and down, select the up arrow or the down arrow in the upper-right corner of the screen, or use the up arrow key or the down arrow key on the keyboard.
- To close open dialog boxes and return to the main Computer Setup screen, press esc, and then follow the on-screen instructions.

To exit Computer Setup, choose one of the following methods:

- To exit Computer Setup menus without saving your changes, select **Main**, select **Ignore Changes and Exit**, and then select **Yes**.

NOTE: If you are using arrow keys to highlight your choice, you must then press enter.

- To save your changes and exit Computer Setup menus, select **Main**, select **Save Changes and Exit**, and then select **Yes**.

NOTE: If you are using arrow keys to highlight your choice, you must then press enter.

Your changes go into effect when the computer restarts.

Restoring factory settings in Computer Setup

To return all settings in Computer Setup to the values that were set at the factory, follow these steps.

NOTE: Restoring defaults will not change the hard drive mode.

1. Start Computer Setup. See Using Computer Setup on page 80.
2. Select **Main**, select **Apply Factory Defaults and Exit**, and then select **Yes**.

NOTE: If you are using arrow keys to highlight your choice, you must then press enter.

NOTE: On select products, the selections might display **Restore Defaults** instead of **Apply Factory Defaults and Exit**.

Your changes go into effect when the computer restarts.

NOTE: Your password settings and security settings are not changed when you restore the factory settings.

Updating the BIOS

Updated versions of the BIOS might be available on the HP website. Most BIOS updates on the HP website are packaged in compressed files called *SoftPaqs*.

Some download packages contain a file named Readme.txt, which contains information regarding installing and troubleshooting the file.

Determining the BIOS version

To decide whether you need to update Computer Setup (BIOS), first determine the BIOS version on your computer.

If you are already in Windows, you can access BIOS version information (also known as *ROM date* and *System BIOS*) by pressing fn+esc (select products only). Or you can use Computer Setup.

1. Start Computer Setup. See Using Computer Setup on page 80.
2. Select **Main**, and then select **System Information**.
3. To exit Computer Setup menus without saving your changes, select **Main**, select **Ignore Changes and Exit**, and then select **Yes**.

NOTE: If you are using arrow keys to highlight your choice, you must then press enter.

To check for later BIOS versions, see Preparing for a BIOS update on page 81.

Preparing for a BIOS update

Be sure to follow all prerequisites before downloading and installing a BIOS update.

IMPORTANT: To reduce the risk of damage to the computer or an unsuccessful installation, download and install a BIOS update only when the computer is connected to reliable external power using the AC adapter. Do not download or install a BIOS update while the computer is running on battery power, docked in an optional docking device, or connected to an optional power source. During the download and installation, follow these instructions:

Do not disconnect power on the computer by unplugging the power cord from the AC outlet.

Do not shut down the computer or initiate Sleep.

Do not insert, remove, connect, or disconnect any device, cable, or cord.

Downloading a BIOS update

After you review the prerequisites, you can check for and download BIOS updates.

1. Select the **Search** icon in the taskbar, type `support` in the search box, and then select the **HP Support Assistant** app.

   – or –

   Select the question mark icon (select products only) in the taskbar.

2. Select **Updates**, and then select **Check for updates and messages**.

3. Follow the on-screen instructions.

4. At the download area, follow these steps:

   a. Identify the most recent BIOS update and compare it to the BIOS version currently installed on your computer. Make a note of the date, name, or other identifier. You might need this information to locate the update later, after it has been downloaded to your hard drive.

   b. Follow the on-screen instructions to download your selection to the hard drive.

      Make a note of the path to the location on your hard drive where the BIOS update is downloaded. You will need to access this path when you are ready to install the update.

NOTE: If you connect your computer to a network, consult the network administrator before installing any software updates, especially system BIOS updates.

Installing a BIOS update

BIOS installation procedures vary. Follow any instructions that are displayed on the screen after the download is complete. If no instructions are displayed, follow these steps.

1. Select the **Search** icon in the taskbar, type `file` in the search box, and then select **File Explorer**.

2. Select your hard drive designation. The hard drive designation is typically Local Disk (C:).

3. Using the hard drive path you recorded earlier, open the folder that contains the update.

4. Double-click the file that has an .exe extension (for example, *filename*.exe).

   The BIOS installation begins.

5. Complete the installation by following the on-screen instructions.

NOTE: After a message on the screen reports a successful installation, you can delete the downloaded file from your hard drive.

Changing the boot order using the f9 prompt

To dynamically choose a boot device for the current startup sequence, follow these steps.

1. Access the Boot Device Options menu:

   - Turn on or restart the computer, and when the HP logo appears, press f9 to enter the Boot Device Options menu.

2. Select a boot device, press enter, and then follow the on-screen instructions.

TPM BIOS settings (select products only)

TPM provides additional security for your computer. You can modify the TPM settings in Computer Setup (BIOS).

IMPORTANT: Before enabling Trusted Platform Module (TPM) functionality on this system, you must ensure that your intended use of TPM complies with relevant local laws, regulations and policies, and approvals or licenses must be obtained if applicable. For any compliance issues arising from your operation or usage of TPM that violates the previously mentioned requirement, you shall bear all the liabilities wholly and solely. HP will not be responsible for any related liabilities.

NOTE: If you change the TPM setting to Hidden, TPM is not visible in the operating system.

To access TPM settings in Computer Setup:

1. Start Computer Setup. See Using Computer Setup on page 80.
2. Select **Security**, select **TPM Embedded Security**, and then follow the on-screen instructions.

Using HP Sure Start (select products only)

Select computer models are configured with HP Sure Start, a technology that monitors the computer's BIOS for attacks or corruption. If the BIOS becomes corrupted or is attacked, HP Sure Start automatically restores the BIOS to its previously safe state, without user intervention.

HP Sure Start is configured and already enabled so that most users can use the HP Sure Start default configuration. Advanced users can customize the default configuration.

To access the latest documentation on HP Sure Start, go to http://www.hp.com/support. Select **Find your product**, and then follow the on-screen instructions.

8 Using HP PC Hardware Diagnostics

You can use the HP PC Hardware Diagnostics utility to determine whether your computer hardware is running properly. The three versions are HP PC Hardware Diagnostics Windows, HP PC Hardware Diagnostics UEFI (Unified Extensible Firmware Interface), and (for select products only) Remote HP PC Hardware Diagnostics UEFI, a firmware feature.

Using HP PC Hardware Diagnostics Windows (select products only)

HP PC Hardware Diagnostics Windows is a Windows-based utility that allows you to run diagnostic tests to determine whether the computer hardware is functioning properly. The tool runs within the Windows operating system to diagnose hardware failures.

If HP PC Hardware Diagnostics Windows is not installed on your computer, you must download and install it. To download HP PC Hardware Diagnostics Windows, see Downloading HP PC Hardware Diagnostics Windows on page 85.

Using an HP PC Hardware Diagnostics Windows hardware failure ID code

When HP PC Hardware Diagnostics Windows detects a failure that requires hardware replacement, a 24-digit failure ID code is generated for select component tests. For interactive tests, such as keyboard, mouse, or audio and video palette, you must perform troubleshooting steps before you can receive a failure ID.

- You have several options after you receive a failure ID:
  - Select **Next** to open the Event Automation Service (EAS) page, where you can log the case.
  - Scan the QR code with your mobile device, which takes you to the EAS page, where you can log the case.
  - Select the box next to the 24-digit failure ID to copy your failure code and send it to support.

Accessing HP PC Hardware Diagnostics Windows

After HP PC Hardware Diagnostics Windows is installed, you can access it from HP Support Assistant or the Start menu.

Accessing HP PC Hardware Diagnostics Windows from HP Support Assistant

After HP PC Hardware Diagnostics Windows is installed, follow these steps to access it from HP Support Assistant:

1. Complete one of the following tasks:
   - Select the **Search** icon in the taskbar, type `support` in the search box, and then select the **HP Support Assistant** app.
   - Select the question mark icon in the taskbar.
2. Select **Fixes & Diagnostics**.
3. Select **Run hardware diagnostics**, and then select **Launch**.

4. When the tool opens, select the type of diagnostic test that you want to run, and then follow the on-screen instructions.

NOTE: To stop a diagnostic test, select **Cancel**.

Accessing HP PC Hardware Diagnostics Windows from the Start menu (select products only)

After HP PC Hardware Diagnostics Windows is installed, follow these steps to access it from the Start menu:

1. Select the **Start** button, and then select **All apps**.
2. Select **HP PC Hardware Diagnostics Windows**.
3. When the tool opens, select the type of diagnostic test that you want to run, and then follow the on-screen instructions.

NOTE: To stop a diagnostic test, select **Cancel**.

Downloading HP PC Hardware Diagnostics Windows

The HP PC Hardware Diagnostics Windows downloading instructions are provided in English only. You must use a Windows computer to download this tool because only .exe files are provided.

Downloading the latest HP PC Hardware Diagnostics Windows version from HP

To download HP PC Hardware Diagnostics Windows from HP, follow these steps:

1. Go to http://www.hp.com/go/techcenter/pcdiags. The HP PC Diagnostics home page is displayed.
2. Select **Download HP Diagnostics Windows**, and then select the specific Windows diagnostics version to download to your computer or a USB flash drive.

The tool downloads to the selected location.

Downloading the HP PC Hardware Diagnostics Windows from the Microsoft Store

You can download the HP PC Hardware Diagnostics Windows from the Microsoft Store:

1. Select the Microsoft Store app on your desktop or select the **Search** icon in the taskbar, and then type `Microsoft Store` in the search box.
2. Type `HP PC Hardware Diagnostics Windows` in the **Microsoft Store** search box.
3. Follow the on-screen directions.

The tool downloads to the selected location.

Downloading HP Hardware Diagnostics Windows by product name or number (select products only)

You can download HP PC Hardware Diagnostics Windows by product name or number.

NOTE: For some products, you might have to download the software to a USB flash drive by using the product name or number.

1. Go to http://www.hp.com/support.
2. Select **Software and Drivers**, select your type of product, and then enter the product name or number in the search box that is displayed.

3. In the **Diagnostics** section, select **Download**, and then follow the on-screen instructions to select the specific Windows diagnostics version to be downloaded to your computer or USB flash drive.

The tool downloads to the selected location.

Installing HP PC Hardware Diagnostics Windows

To install HP PC Hardware Diagnostics Windows, navigate to the folder on your computer or the USB flash drive where the .exe file downloaded, double-click the .exe file, and then follow the on-screen instructions.

Using HP PC Hardware Diagnostics UEFI

HP PC Hardware Diagnostics UEFI (Unified Extensible Firmware Interface) allows you to run diagnostic tests to determine whether the computer hardware is functioning properly. The tool runs outside the operating system so that it can isolate hardware failures from issues that are caused by the operating system or other software components.

NOTE: For some products, you must use a Windows computer and a USB flash drive to download and create the HP UEFI support environment because only .exe files are provided. For more information, see Downloading HP PC Hardware Diagnostics UEFI to a USB flash drive on page 87.

If your PC does not start in Windows, you can use HP PC Hardware Diagnostics UEFI to diagnose hardware issues.

Using an HP PC Hardware Diagnostics UEFI hardware failure ID code

When HP PC Hardware Diagnostics UEFI detects a failure that requires hardware replacement, a 24-digit failure ID code is generated.

For assistance in solving the problem:

- Select **Contact HP**, accept the HP privacy disclaimer, and then use a mobile device to scan the failure ID code that appears on the next screen. The HP Customer Support - Service Center page appears with your failure ID and product number automatically filled in. Follow the on-screen instructions.

  – or –

  Contact support, and provide the failure ID code.

NOTE: To start diagnostics on a convertible computer, your computer must be in notebook mode, and you must use the attached keyboard.

NOTE: If you need to stop a diagnostic test, press esc.

Starting HP PC Hardware Diagnostics UEFI

To start HP PC Hardware Diagnostics UEFI, follow this procedure.

1. Turn on or restart the computer, and quickly press esc.
2. Press f2.

   The BIOS searches three places for the diagnostic tools, in the following order:

a. Connected USB flash drive

NOTE: To download the HP PC Hardware Diagnostics UEFI tool to a USB flash drive, see Downloading the latest HP PC Hardware Diagnostics UEFI version on page 87.

b. Hard drive

c. BIOS

3. When the diagnostic tool opens, select the type of diagnostic test that you want to run, and then follow the on-screen instructions.

Downloading HP PC Hardware Diagnostics UEFI to a USB flash drive

Downloading HP PC Hardware Diagnostics UEFI to a USB flash drive can be useful in some situations.

- HP PC Hardware Diagnostics UEFI is not included in the preinstallation image.
- HP PC Hardware Diagnostics UEFI is not included in the HP Tool partition.
- The hard drive is damaged.

NOTE: The HP PC Hardware Diagnostics UEFI downloading instructions are provided in English only, and you must use a Windows computer to download and create the HP UEFI support environment because only .exe files are provided.

Downloading the latest HP PC Hardware Diagnostics UEFI version

To download the latest HP PC Hardware Diagnostics UEFI version to a USB flash drive, follow this procedure:

1. Go to http://www.hp.com/go/techcenter/pcdiags. The HP PC Diagnostics home page is displayed.
2. Select **Download HP Diagnostics UEFI**, and then select **Run**.

Downloading HP PC Hardware Diagnostics UEFI by product name or number (select products only)

You can download HP PC Hardware Diagnostics UEFI by product name or number (select products only) to a USB flash drive.

NOTE: For some products, you might have to download the software to a USB flash drive by using the product name or number.

1. Go to http://www.hp.com/support.
2. Enter the product name or number, select your computer, and then select your operating system.
3. In the **Diagnostics** section, follow the on-screen instructions to select and download the specific UEFI Diagnostics version for your computer.

Using Remote HP PC Hardware Diagnostics UEFI settings (select products only)

Remote HP PC Hardware Diagnostics UEFI is a firmware (BIOS) feature that downloads HP PC Hardware Diagnostics UEFI to your computer. It can then execute the diagnostics on your computer, and it might upload results to a preconfigured server.

For more information about Remote HP PC Hardware Diagnostics UEFI, go to http://www.hp.com/go/techcenter/pcdiags, and then select **Find out more**.

Downloading Remote HP PC Hardware Diagnostics UEFI

HP Remote PC Hardware Diagnostics UEFI is also available as a SoftPaq that you can download to a server.

Downloading the latest Remote HP PC Hardware Diagnostics UEFI version

You can download the latest HP PC Hardware Diagnostics UEFI version to a USB flash drive.

1. Go to http://www.hp.com/go/techcenter/pcdiags. The HP PC Diagnostics home page is displayed.
2. Select **Download Remote Diagnostics**, and then select **Run**.

Downloading Remote HP PC Hardware Diagnostics UEFI by product name or number

You can download HP Remote PC Hardware Diagnostics UEFI by product name or number.

NOTE: For some products, you might have to download the software by using the product name or number.

1. Go to http://www.hp.com/support.
2. Select **Software and Drivers**, select your type of product, enter the product name or number in the search box that is displayed, select your computer, and then select your operating system.
3. In the **Diagnostics** section, follow the on-screen instructions to select and download the **Remote UEFI** version for the product.

Customizing Remote HP PC Hardware Diagnostics UEFI settings

Using the Remote HP PC Hardware Diagnostics setting in Computer Setup (BIOS), you can perform several customizations.

- Set a schedule for running diagnostics unattended. You can also start diagnostics immediately in interactive mode by selecting **Execute Remote HP PC Hardware Diagnostics**.
- Set the location for downloading the diagnostic tools. This feature provides access to the tools from the HP website or from a server that has been preconfigured for use. Your computer does not require the traditional local storage (such as a hard drive or USB flash drive) to run remote diagnostics.
- Set a location for storing the test results. You can also set the user name and password that you use for uploads.
- Display status information about the diagnostics run previously.

To customize Remote HP PC Hardware Diagnostics UEFI settings, follow these steps:

1. Turn on or restart the computer, and when the HP logo appears, press f10 to enter Computer Setup.
2. Select **Advanced**, and then select **Settings**.
3. Make your customization selections.
4. Select **Exit**, then select **Save Changes and Exit** to save your settings.

Your changes take effect when the computer restarts.

9 Specifications

This chapter provides specifications for your computer system.

Computer specifications

This section provides specifications for your computer. When traveling with your computer, the computer dimensions and weights, as well as input power ratings and operating specifications, provide helpful information.

Table 9-1 Computer specifications

	Metric	U.S.
Dimensions		
Width	**296.4 mm**	11.67 in
Depth	**234.3 mm**	9.22 in
Height	**17.9 mm**	0.70 in
Weight (starting)	**1.40 kg**	3.09 lbs
Input power		
Operating voltage and current	19.5 V dc @ 3.33 A – 65 W	
Temperature		
Operating	**5°C to 35°C**	41°F to 95°F
Nonoperating	**–20°C to 60°C**	–4°F to 140°F
Relative humidity (noncondensing)		
Operating	10% to 90%	
Nonoperating	5% to 95%	
Maximum altitude (unpressurized)		
Operating	**–15 m to 3,048 m**	–50 ft to 10,000 ft
Nonoperating	**–15 m to 12,192 m**	–50 ft to 40,000 ft

NOTE: Applicable product safety standards specify thermal limits for plastic surfaces. The device operates well within this range of temperatures.

Display specifications

This section provides specifications for your display.

Table 9-2 Display specifications

	Metric	U.S.
Active diagonal size	**34.3 cm**	13.5 in

Table 9-2 Display specifications (continued)

	Metric	U.S.
Resolution	3000 × 2000 (3k2k) 1920 × 1280 (WUXGA+)	
Surface treatment	BrightView Antiglare	
Brightness	400 nits 1000 nits (privacy panel)	
Viewing angle	UWVA	
Backlight	WLED	

Solid-state drive specifications

This section provides specifications for your solid-state drives.

Table 9-3 Solid-state drive specifications

	256 GB*	512 GB*	1 TB*	2 TB*
Dimensions				
Height	1.0 mm	1.0 mm	1.0 mm	1.0 mm
Length	50.8 mm	50.8 mm	50.8 mm	50.8 mm
Width	28.9 mm	28.9 mm	28.9 mm	29.0 mm
Weight	< 10 g	< 10 g	< 10 g	< 10 g
Interface type	PCIe	PCIe	PCIe	PCIe
Ready time, maximum (to not busy)	1.0 ms	< 1.0 ms	1.0 ms	1.0 ms
Access times, logical	0.1 ms	0.1 ms	0.1 ms	0.1 ms
Transfer rate				
Sequential read	up to 2150 MBps	up to 2150 MBps	up to 2150 MBps	up to 2150 MBps
Random read	Up to 300,000 IOPs	Up to 300,000 IOPs	Up to 300,000 IOPs	Up to 300,000 IOPs
Sequential write	up to 1550 MBps	up to 1550 MBps	up to 1550 MBps	up to 1550 MBps
Random write	Up to 100,000 IOPs	Up to 100,000 IOPs	Up to 100,000 IOPs	Up to 100,000 IOPs
Total logical sectors	468,883,296	1,000,215,216	1,500,336,388	2,988,316,776
Operating temperature	**0°C to 70°C** (32°F to 158°F)			

*1 GB = 1 billion bytes when referring to hard drive storage capacity. Actual accessible capacity is less. Actual drive specifications may differ slightly.

NOTE: Certain restrictions and exclusions apply. Contact support for details.

10 Statement of memory volatility

For general information regarding nonvolatile memory in HP business computers, and to restore nonvolatile memory that can contain personal data after the system has been turned off and the hard drive has been removed, use these instructions.

HP business computer products that use Intel®-based or AMD®-based system boards contain volatile DDR memory. The amount of nonvolatile memory present in the system depends upon the system configuration. Intel-based and AMD-based system boards contain nonvolatile memory subcomponents as originally shipped from HP, with the following assumptions:

- No subsequent modifications were made to the system.
- No applications, features, or functionality were added to or installed on the system.

Following system shutdown and removal of all power sources from an HP business computer system, personal data can remain on volatile system memory (DIMMs) for a finite period of time and also remains in nonvolatile memory. Use the following steps to remove personal data from the computer, including the nonvolatile memory found in Intel-based and AMD-based system boards.

NOTE: If your tablet has a keyboard base, connect to the keyboard base before beginning steps in this chapter.

Current BIOS steps

Use these instructions to restore nonvolatile memory.

1. Follow these steps to restore the nonvolatile memory that can contain personal data. Restoring or reprogramming nonvolatile memory that does not store personal data is neither necessary nor recommended.

   a. Turn on or restart the computer, and then quickly press esc.

   NOTE: If the system has a BIOS administrator password, type the password at the prompt.

   b. Select **Main**, select **Apply Factory Defaults and Exit**, and then select **Yes** to load defaults. The computer restarts.

   c. During the restart, press esc while the "Press the ESC key for Startup Menu" message is displayed at the bottom of the screen.

   NOTE: If the system has a BIOS administrator password, type the password at the prompt.

   d. Select the **Security** menu, select **Restore Security Settings to Factory Defaults**, and then select **Yes** to restore security level defaults. The computer reboots.

   e. During the reboot, press esc while the "Press the ESC key for Startup Menu" message is displayed at the bottom of the screen.

   NOTE: If the system has a BIOS administrator password, type the password at the prompt.

f. If an asset or ownership tag is set, select the **Security** menu and scroll down to the **Utilities** menu. Select **System IDs**, and then select **Asset Tracking Number**. Clear the tag, and then make the selection to return to the prior menu.

g. If a DriveLock password is set, select the **Security** menu, and scroll down to **Hard Drive Utilities** under the **Utilities** menu. Select **Hard Drive Utilities**, select **DriveLock**, and then clear the check box for **DriveLock password on restart**. Select **OK** to proceed.

h. Select the **Main** menu, and then select **Reset BIOS Security to factory default**. Select **Yes** at the warning message. The computer reboots.

i. During the reboot, press esc while the "Press the ESC key for Startup Menu" message is displayed at the bottom of the screen.

NOTE: If the system has a BIOS administrator password, type the password at the prompt.

j. Select the **Main** menu, select **Apply Factory Defaults and Exit**, select **Yes** to save changes and exit, and then select **Shutdown**.

k. Reboot the system. If the system has a Trusted Platform Module (TPM), fingerprint reader, or both, one or two prompts will appear—one to clear the TPM and the other to Reset Fingerprint Sensor. Press or tap f1 to accept or f2 to reject.

l. Remove all power and system batteries for at least 24 hours.

2. Complete one of the following:

- Remove and retain the storage drive.

  – or –

- Clear the drive contents by using a third-party utility designed to erase data from an SSD.

  – or –

- Clear the contents of the drive by using the following BIOS Setup Secure Erase command option steps:

NOTE: If you clear data using Secure Erase, you cannot recover it.

a. Turn on or restart the computer, and then quickly press esc.

b. Select the **Security** menu and scroll down to the esc menu.

c. Select **Hard Drive Utilities**.

d. Under **Utilities**, select **Secure Erase**, select the hard drive storing the data you want to clear, and then follow the on-screen instructions to continue.

   – or –

   Clear the contents of the drive using the following Disk Sanitizer commands steps:

   i. Turn on or restart the computer, and then quickly press esc.

   ii. Select the **Security** menu and scroll down to the **Utilities** menu.

   iii. Select **Hard Drive Utilities**.

iv. Under **Utilities**, select **Disk Sanitizer**, select the hard drive with the data that you want to clear, and then follow the on-screen instructions to continue.

NOTE: The amount of time it takes for Disk Sanitizer to run can take several hours. Plug the computer into an AC outlet before starting.

Nonvolatile memory usage

Use this table to troubleshoot nonvolatile memory usage.

Table 10-1 Troubleshooting steps for nonvolatile memory usage

Nonvolatile memory type	Amount (Size)	Does this memory store customer data?	Does this memory retain data when power is removed?	What is the purpose of this memory?	How is data entered into this memory?	How is this memory write-protected?
HP Sure Start flash (select models only)	8 MB	No	Yes	Provides protected backup of critical System BIOS code, EC firmware, and critical computer configuration data for select platforms that support HP Sure Start. For more information, see Using HP Sure Start (select models only)HP Sure Start on page 97.	Data cannot be written to this device via the host processor. The content is managed solely by the HP Sure Start Embedded Controller.	This memory is protected by the HP Sure Start Embedded Controller.
Real Time Clock (RTC) battery backed-up CMOS configuration memory	256 bytes	No	Yes	Stores system date and time and noncritical data.	RTC battery backed-up CMOS is programmed using Computer Setup (BIOS), or by changing the Windows date & time.	This memory is not write-protected.
Controller (NIC) EEPROM	64 KB (not customer accessible)	No	Yes	Stores NIC configuration and NIC firmware.	NIC EEPROM is programmed using a utility from the NIC vendor that can be run from DOS.	A utility must be used to write data to this memory and is available from the NIC vendor. Writing data to this ROM in an inappropriate manner will render the NIC nonfunctional.

Table 10-1 Troubleshooting steps for nonvolatile memory usage (continued)

Nonvolatile memory type	Amount (Size)	Does this memory store customer data?	Does this memory retain data when power is removed?	What is the purpose of this memory?	How is data entered into this memory?	How is this memory write-protected?
DIMM Serial Presence Detect (SPD) configuration data	256 bytes per memory module, 128 bytes programmable (not customer accessible)	No	Yes	Stores memory module information.	DIMM SPD is programmed by the memory vendor.	Data cannot be written to this memory when the module is installed in a computer. The specific write-protection method varies by memory vendor.
System BIOS	9 MB	Yes	Yes	Stores system BIOS code and computer configuration data.	System BIOS code is programmed at the factory. Code is updated when the system BIOS is updated. Configuration data and settings are entered using the Computer Setup (BIOS) or a custom utility.	**NOTE:** Writing data to this ROM in an inappropriate manner can render the computer nonfunctional. A utility must be used for writing data to this memory and is available on the HP website; go to http://www.hp.com/support. Select **Find your product**, and then follow the on-screen instructions.
Intel Management Engine Firmware (present only in select Elite or Z models. For more information, go to http://www.hp.com/support. Select **Identify your product for manuals and specific product information**, and then follow the on-screen instructions.)	1.5 MB or 7 MB	Yes	Yes	Stores Management Engine Code, Settings, Provisioning Data and iAMT third-party data store.	Management Engine Code is programmed at the factory. Code is updated via Intel secure firmware update utility. Unique Provisioning Data can be entered at the factory or by an administrator using the Management Engine (MEBx) setup utility. The third-party data store contents can be populated by a remote management console or local applications that have been registered by an administrator to have access to the space.	The Intel chipset is configured to enforce hardware protection to block all direct read-write access to this area. An Intel utility must be used for updating the firmware. Only firmware updates digitally signed by Intel can be applied using this utility.
Bluetooth flash (select products only)	2 megabits	No	Yes	Stores Bluetooth configuration and firmware.	Bluetooth flash is programmed at the factory. Tools for writing data to this memory are not publicly available but can be obtained from the silicon vendor.	A utility must be used for writing data to this memory and is made available through newer versions of the driver whenever the flash requires an upgrade.

Table 10-1 Troubleshooting steps for nonvolatile memory usage (continued)

Nonvolatile memory type	Amount (Size)	Does this memory store customer data?	Does this memory retain data when power is removed?	What is the purpose of this memory?	How is data entered into this memory?	How is this memory write-protected?
802.11 WLAN EEPROM	4 kilobits to 8 kilobits	No	Yes	Stores configuration and calibration data.	802.11 WLAN EEPROM is programmed at the factory. Tools for writing data to this memory are not made public.	A utility must be used for writing data to this memory and is typically not made available to the public unless a firmware upgrade is necessary to address a unique issue.
Camera (select products only)	64 kilobits	No	Yes	Stores camera configuration and firmware.	Camera memory is programmed using a utility from the device manufacturer that can be run from Windows.	A utility must be used for writing data to this memory and is typically not made available to the public unless a firmware upgrade is necessary to address a unique issue.
Fingerprint reader (select products only)	512 KB flash	Yes	Yes	Stores fingerprint templates.	Fingerprint reader memory is programmed by user enrollment in HP ProtectTools Security Manager.	Only a digitally signed application can make the call to write to the flash.

Questions and answers

Use this section to answer your questions about nonvolatile memory.

1. **How can the BIOS settings be restored (returned to factory settings)?**

IMPORTANT: The restore defaults feature does not securely erase any information on your hard drive. See question and answer 6 for steps to securely erase information.

The restore defaults feature does not reset the Custom Secure Boot keys. See question and answer 7 for information about resetting the keys.

a. Turn on or restart the computer, and then quickly press esc.

b. Select **Main**, and then select **Apply Factory Defaults and Exit**.

c. Follow the on-screen instructions.

d. Select **Main**, select **Save Changes and Exit**, and then follow the on-screen instructions.

2. **What is a UEFI BIOS, and how is it different from a legacy BIOS?**

The Unified Extensible Firmware Interface (UEFI) BIOS is an industry-standard software interface between the platform firmware and an operating system (OS). It replaces the older BIOS architecture but supports much of the legacy BIOS functionality.

Like the legacy BIOS, the UEFI BIOS provides an interface to display the system information and configuration settings and to change the configuration of your computer before an OS is loaded. BIOS provides a secure runtime environment that supports a Graphic User Interface (GUI). In this environment, you can use either a pointing device (touch screen, touchpad, pointing stick, or USB mouse) or the keyboard to navigate and make menu and configuration selections. The UEFI BIOS also contains basic system diagnostics.

The UEFI BIOS provides functionality beyond that of the legacy BIOS. In addition, the UEFI BIOS works to initialize the computer's hardware before loading and executing the OS; the runtime environment allows the loading and execution of software programs from storage devices to provide more functionality, such as advanced hardware diagnostics (with the ability to display more detailed system information) and advanced firmware management and recovery software.

HP has provided options in Computer Setup (BIOS) to allow you to run in legacy BIOS, if required by the operating system. Examples of this requirement would be if you upgrade or downgrade the OS.

3. **Where is the UEFI BIOS located?**

   The UEFI BIOS is located on a flash memory chip. You must use a utility to write to the chip.

4. **What kind of configuration data is stored on the DIMM Serial Presence Detect (SPD) memory module? How would this data be written?**

   The DIMM SPD memory contains information about the memory module, such as size, serial number, data width, speed and timing, voltage, and thermal information. This information is written by the module manufacturer and stored on an EEPROM. You cannot write to this EEPROM when the memory module is installed in a computer. Third-party tools do exist that can write to the EEPROM when the memory module is not installed in a computer. Various third-party tools are available to read SPD memory.

5. **What is meant by "Restore the nonvolatile memory found in Intel-based system boards"?**

   This message relates to clearing the Real Time Clock (RTC) CMOS memory that contains computer configuration data.

6. **How can the BIOS security be reset to factory defaults and erase the data?**

   IMPORTANT: Resetting results in the loss of information.

   These steps do not reset Custom Secure Boot Keys. See question and answer 7 for information about resetting the keys.

   a. Turn on or restart the computer, and then quickly press esc.

   b. Select **Main**, and then select **Reset Security to Factory Defaults**.

   c. Follow the on-screen instructions.

   d. Select **Main**, select **Save Changes and Exit**, and then follow the on-screen instructions.

7. **How can the Custom Secure Boot Keys be reset?**

   Secure Boot is a feature to ensure that only authenticated code can start on a platform. If you enabled Secure Boot and created Custom Secure Boot Keys, disabling Secure Boot does not clear the keys. You must also select to clear the Custom Secure Boot Keys. Use the same Secure Boot access procedure that you used to create the Custom Secure Boot Keys, but select to clear or delete all Secure Boot Keys.

   a. Turn on or restart the computer, and then quickly press esc.

b. Select the **Security** menu, select **Secure Boot Configuration**, and then follow the on-screen instructions.

c. At the **Secure Boot Configuration** window, select **Secure Boot**, select **Clear Secure Boot Keys**, and then follow the on-screen instructions to continue.

Using HP Sure Start (select products only)

Select computer models are configured with HP Sure Start, a technology that continuously monitors your computer's BIOS for attacks or corruption.

If the BIOS becomes corrupted or is attacked, HP Sure Start restores the BIOS to its previously safe state, without user intervention. Those select computer models ship with HP Sure Start configured and enabled. HP Sure Start is configured and already enabled so that most users can use the HP Sure Start default configuration. Advanced users can customize the default configuration.

To access the latest documentation on HP Sure Start, go to http://www.hp.com/support.

11 Recycling

When a nonrechargeable or rechargeable battery has reached the end of its useful life, do not dispose of the battery in general household waste. Follow the local laws and regulations in your area for battery disposal.

HP encourages customers to recycle used electronic hardware, HP original print cartridges, and rechargeable batteries. For more information about recycling programs, see the HP website at http://www.hp.com/recycle.

12 Power cord set requirements

This chapter provides power cord requirements for countries and regions.

The wide-range input feature of the computer permits it to operate from any line voltage from 100 V AC to 120 V AC, or from 220 V AC to 240 V AC.

The three-conductor power cord set included with the computer meets the requirements for use in the country or region where the equipment is purchased.

Power cord sets for use in other countries or regions must meet the requirements of the country and region where the computer is used.

Requirements for all countries

These power cord requirements are applicable to all countries and regions.

- The length of the power cord set must be at least **1.0 m** (3.3 ft) and no more than **2.0 m** (6.5 ft).
- All power cord sets must be approved by an acceptable accredited agency responsible for evaluation in the country or region where the power cord set will be used.
- The power cord sets must have a minimum current capacity of 10 A and a nominal voltage rating of 125 V ac or 250 V ac, as required by the power system of each country or region.
- The appliance coupler must meet the mechanical configuration of an EN 60 320/IEC 320 Standard Sheet C13 connector for mating with the appliance inlet on the back of the computer.

Requirements for specific countries and regions

To determine power cord requirements for specific countries and regions, use this table.

Table 12-1 Power cord requirements for specific countries and regions

Country/region	Accredited agency	Applicable note number
Argentina	IRAM	1
Australia	SAA	1
Austria	OVE	1
Belgium	CEBEC	1
Brazil	ABNT	1
Canada	CSA	2
Chile	IMQ	1
Denmark	DEMKO	1
Finland	FIMKO	1
France	UTE	1
Germany	VDE	1

Table 12-1 Power cord requirements for specific countries and regions (continued)

Country/region	Accredited agency	Applicable note number
India	BIS	1
Israel	SII	1
Italy	IMQ	1
Japan	JIS	3
Netherlands	KEMA	1
New Zealand	SANZ	1
Norway	NEMKO	1
People's Republic of China	CCC	4
Saudi Arabia	SASO	7
Singapore	PSB	1
South Africa	SABS	1
South Korea	KTL	5
Sweden	SEMKO	1
Switzerland	SEV	1
Taiwan	BSMI	6
Thailand	TISI	1
United Kingdom	ASTA	1
United States	UL	2

1. The flexible cord must be Type H05VV-F, three-conductor, 0.75 mm² conductor size. Power cord set fittings (appliance coupler and wall plug) must bear the certification mark of the agency responsible for evaluation in the country or region where it will be used.
2. The flexible cord must be Type SVT/SJT or equivalent, No. 18 AWG, three-conductor. The wall plug must be a two-pole grounding type with a NEMA 5-15P (15 A, 125 V ac) or NEMA 6-15P (15 A, 250 V ac) configuration. CSA or C-UL mark. UL file number must be on each element.
3. The appliance coupler, flexible cord, and wall plug must bear a T mark and registration number in accordance with the Japanese Dentori Law. The flexible cord must be Type VCTF, three-conductor, 0.75 mm² or 1.25 mm² conductor size. The wall plug must be a two-pole grounding type with a Japanese Industrial Standard C8303 (7 A, 125 V ac) configuration.
4. The flexible cord must be Type RVV, three-conductor, 0.75 mm² conductor size. Power cord set fittings (appliance coupler and wall plug) must bear the CCC certification mark.
5. The flexible cord must be Type H05VV-F three-conductor, 0.75 mm² conductor size. KTL logo and individual approval number must be on each element. Approval number and logo must be printed on a flag label.
6. The flexible cord must be Type HVCTF three-conductor, 1.25 mm² conductor size. Power cord set fittings (appliance coupler, cable, and wall plug) must bear the BSMI certification mark.
7. For 127 V ac, the flexible cord must be Type SVT or SJT 3-conductor, 18 AWG, with plug NEMA 5-15P (15 A, 125 V ac), with UL and CSA or C-UL marks. For 240 V ac, the flexible cord must be Type H05VV-F three-conductor, 0.75 mm² or 1.00 mm² conductor size, with plug BS 1363/A with BSI or ASTA marks.

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